

WEYL–CHAMBER GEOMETRY, POINCARÉ INEQUALITY MEETS SCHATTEN CLASS PROPERTIES OF DUNKL–RIESZ TRANSFORM COMMUTATORS

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ABSTRACT.

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1. INTRODUCTION

Definition 1.1. Suppose $1 < p < \infty$. We say that a function $f \in L^p_{\text{loc}}(\mathbb{R}^N, dw)$ belongs to Dunkl Besov space $B^p_{\text{Dunkl}}(\mathbb{R}^N)$ if

$$\|f\|_{B^p_{\text{Dunkl}}(\mathbb{R}^N)} := \left(\int_{\mathbb{R}^N} \int_{\mathbb{R}^N} \left(\frac{d(x, y)}{\|x - y\|} \right)^p \frac{|f(x) - f(y)|^p}{w(B(x, d(x, y)))^2} dw(x) dw(y) \right)^{1/p} < +\infty.$$

1.1. Notation. Conventionally, we let $\mathbb{N}$ denote the set of positive integers and set $\mathbb{Z}_+ := \{0\} \cup \mathbb{N}$. Throughout the paper, the symbols C and c denote positive constants (possibly varying from line to line) that may depend only on N , p , and the structural data (R, k) of the Dunkl setting, but are independent of the main parameters (e.g., the symbol b and auxiliary decay factors). We write $A \lesssim B$ to mean that $A \leq CB$ for some constant $C > 0$, and $A \sim B$ to mean that both $A \lesssim B$ and $B \lesssim A$ hold. For $\lambda > 0$ and a cube Q with sidelength $\ell(Q)$, write λQ for the cube concentric with Q and sidelength $\lambda \ell(Q)$. Similarly, for every ball $B = B(x, r)$, set $\lambda B := B(x, \lambda r)$. For any $1 \leq p \leq \infty$, we denote by p' the conjugate of p , which satisfies $\frac{1}{p} + \frac{1}{p'} = 1$. We denote the average of a function f over a μ -measurable set E by

$$(1.1) \quad (f)_E := \int_E f(x) d\mu(x) := \frac{1}{\mu(E)} \int_E f(x) d\mu(x).$$

For a subset $E \subset \mathbb{R}^N$, we denote by χ_E its characteristic function.

Denote by $\{e_j\}_{j=1}^N$ the standard orthonormal basis on $\mathbb{R}^N$.

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2. PRELIMINARIES

2.1. Harmonic analysis in the Dunkl setting. Let $\mathbb{R}^N$ be the Euclidean space equipped with the usual scalar product $\langle \cdot, \cdot \rangle$ and the induced norm $\|\cdot\|$. For a nonzero vector $v \in \mathbb{R}^N$, the reflection σ_v with respect to the hyperplane $v^\perp$ orthogonal to v is defined by

$$\sigma_v x := x - 2 \frac{\langle x, v \rangle}{\|v\|^2} v.$$

A finite set $R \subset \mathbb{R}^N \setminus \{0\}$ is called a root system if $\sigma_v(R) = R$ for every $v \in R$. A root system R is called a normalized reduced root system if $\|v\|^2 = 2$ for every $v \in R$. The finite group G generated by the reflections $\{\sigma_v\}_{v \in R}$ is called the Weyl group (reflection group) of the root system R . Let $k : R \rightarrow [0, \infty)$ be a multiplicity function, which is a G -invariant function fixed throughout the paper. We will denote by $O(x) = \{\sigma(x) : \sigma \in G\}$ the G -orbit of a point $x \in \mathbb{R}^N$, and $O(E) = \bigcup_{x \in E} O(x)$ the G -orbit of $E \subset \mathbb{R}^N$.

Define the Dunkl measure dw on $\mathbb{R}^N$ by

$$dw(x) := \prod_{v \in R} |\langle x, v \rangle|^{k(v)} dx,$$

where dx stands for the Lebesgue measure on $\mathbb{R}^N$. Let $B(x, r) := \{y \in \mathbb{R}^N : \|x - y\| < r\}$ be the Euclidean ball centred at x with radius $r > 0$. Let $\mathbf{N} := N + \sum_{v \in R} k(v)$. Note that

$$w(B(tx, tr)) = t^{\mathbf{N}} w(B(x, r)), \text{ for } x \in \mathbb{R}^N, t, r > 0.$$

Furthermore,

$$(2.1) \quad w(B(x, r)) \sim r^{\mathbf{N}} \prod_{v \in R} (|\langle x, v \rangle| + r)^{k(v)},$$

where the implicit constant only depends on N, R and k . It follows from (2.1) that there exists a constant $C \geq 1$ such that for every $x \in \mathbb{R}^N$,

$$(2.2) \quad C^{-1} \left(\frac{r_2}{r_1} \right)^{\mathbf{N}} \leq \frac{w(B(x, r_2))}{w(B(x, r_1))} \leq C \left(\frac{r_2}{r_1} \right)^{\mathbf{N}}, \text{ for } 0 < r_1 < r_2,$$

so the numbers $\mathbf{N}$ and N are called the upper and lower homogeneous dimensions, respectively.

The Dunkl operators T_ξ are defined by

$$T_\xi f(x) := \partial_\xi f(x) + \sum_{v \in R} \frac{k(v)}{2} \langle v, \xi \rangle \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle},$$

where $\partial_\xi f$ denotes the directional derivative of f in the direction ξ . Furthermore, the Dunkl Laplacian is defined by

$$\Delta_{\text{Dunkl}} f(x) := \sum_{j=1}^N T_{e_j}^2 f(x) = \Delta_{\mathbb{R}^N} f(x) + \sum_{v \in R} k(v) \delta_v f(x),$$

where $\Delta_{\mathbb{R}^N}$ is the Laplacian on $\mathbb{R}^N$ and δ_v is defined by

$$\delta_v f(x) := \frac{\partial_v f(x)}{\langle v, x \rangle} - \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle^2}.$$

In this article, we consider the Dunkl-Riesz transforms defined by $R_{\text{Dunkl},j} := T_{e_j}(-\Delta_{\text{Dunkl}})^{-1/2}$, $j = 1, 2, \dots, N$.

Definition 2.1. The Weyl chambers are defined as the closures of connected components of the set

$$(2.3) \quad \{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \text{ for all } v \in R\}.$$

Define $d(x, y) := \min_{\sigma \in G} \|x - \sigma(y)\|$ as the distance between two G -orbits $O(x)$ and $O(y)$. Recall from [7, Chapter VII, proof of Theorem 2.12] that

$$(2.4) \quad d(x, y) = \|x - y\| \text{ if and only if } x, y \in \Gamma$$

for some closed Weyl chamber Γ .

It is clear that $O(B(x, r)) = \{y \in \mathbb{R}^N : d(x, y) < r\}$. Moreover,

$$w(B(x, r)) \leq w(O(B(x, r))) \leq |G|w(B(x, r))$$

for all $x \in \mathbb{R}^N$ and $r > 0$. We also denote $B^d(x, r) := \{y \in \mathbb{R}^N : d(x, y) < r\}$, i.e., $B^d(x, r) = O(B(x, r))$.

Denote by $K_{\text{Dunkl},j}(x, y)$ the integral kernel of $R_{\text{Dunkl},j}$. We recall the kernel estimates for the Riesz transforms in the Dunkl setting proved in [5]. Although these estimates may not be sharp (see the comment following Theorem 2.1 in [5, p. 10]), they suffice for our purposes.

Lemma 2.2. *There exists a constant C such that for $j \in \{1, 2, \dots, N\}$ and every x, y with $d(x, y) \neq 0$,*

$$|K_{\text{Dunkl},j}(x, y)| \leq C \frac{d(x, y)}{\|x - y\|} \frac{1}{w(B(x, d(x, y)))}.$$

Proof. The proof is given in [5, Theorem 1.1]. □

Lemma 2.3. *For $j \in \{1, 2, \dots, N\}$, and for every ball $B = B(x_0, r) \subset \mathbb{R}^N$, there is another ball $\widehat{B} = B(y_0, r)$ with $y_0 = x_0 + 5re_j$ such that for every $(x, y) \in B \times \widehat{B}$, $K_{\text{Dunkl},j}(x, y)$ does not change sign and satisfies*

$$|K_{\text{Dunkl},j}(x, y)| \geq \frac{C}{w(B(x_0, r))}.$$

Proof. The lower bound is given in [5, Theorem 1.2], while the fact that $K_{\text{Dunkl},j}(x, y)$ does not change sign can be seen in the proof of [5, Theorem 1.2]. Indeed, with the choice $y_0 := x_0 + 5re_j$, we have $y_j - x_j \geq 3r$ for all $(x, y) \in B \times \widehat{B}$. Together with the fact that the heat kernel $h_t(x, y)$ associated with the Dunkl Laplacian is positive, the representation

$$K_{\text{Dunkl},j}(x, y) = -\frac{C}{2} \int_0^\infty \frac{y_j - x_j}{t} h_t(x, y) \frac{dt}{\sqrt{t}},$$

implies that $K_{\text{Dunkl},j}(x, y) < 0$ (and thus, does not change sign) on $B \times \widehat{B}$. □

2.2. Systems of dyadic cubes.

Definition 2.4. Let $X \subseteq \mathbb{R}^N$ be a nonempty open set. A countable family $\mathcal{D} := \bigcup_{k \in \mathbb{Z}} \mathcal{D}_k$, $\mathcal{D}_k := \{Q_\alpha^k : \alpha \in \mathcal{A}_k\}$, of Borel sets $Q_\alpha^k \subseteq X$ is called a *system of dyadic cubes over X* if it has the following properties:

(i) For each $k \in \mathbb{Z}$, we have

$$X = \bigcup_{\alpha \in \mathcal{A}_k} Q_\alpha^k;$$

(ii) If $\ell \geq k$, then either $Q_\beta^\ell \subseteq Q_\alpha^k$ or $Q_\alpha^k \cap Q_\beta^\ell = \emptyset$;

(iii) For each (k, α) and each $\ell \leq k$, there exists a unique β such that $Q_\alpha^k \subseteq Q_\beta^\ell$;

(iv) For each (k, α) there exist 2^N indices β such that

$$Q_\beta^{k+1} \subseteq Q_\alpha^k, \text{ and } Q_\alpha^k = \bigcup_{\substack{Q \in \mathcal{D}_{k+1} \\ Q \subseteq Q_\alpha^k}} Q;$$

(v) For each (k, α) , one has

$$B_X(x_\alpha^k, A_1 2^{-k}) \subseteq Q_\alpha^k \subseteq B_X(x_\alpha^k, A_2 2^{-k}) =: B(Q_\alpha^k)$$

for some positive constants A_1, A_2 depending only on the geometry of X , where we set $B_X(x, r) := B(x, r) \cap X$ for all $x \in X$ and $r > 0$.

(vi) If $\ell \geq k$ and $Q_\beta^\ell \subseteq Q_\alpha^k$, then

$$B(Q_\beta^\ell) \subseteq B(Q_\alpha^k).$$

Each Q_α^k is called a *dyadic cube of generation k* , with center $x_\alpha^k \in Q_\alpha^k$ and side-length $\ell(Q_\alpha^k) = 2^{-k}$.

Notation 2.5. For every dyadic cube $Q \in \mathcal{D}$, we denote by c_Q and $\ell(Q)$ its center and side-length, respectively.

Notation 2.6. Let

$$\mathbb{D}^0 := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k^0$$

be the standard system of dyadic cubes over $\mathbb{R}^N$. Here each dyadic cube is a product of half-open intervals

$$[m_1 2^{-k}, (m_1 + 1) 2^{-k}) \times \cdots \times [m_N 2^{-k}, (m_N + 1) 2^{-k}),$$

with $(m_1, \dots, m_N) \in \mathbb{Z}^N$. Moreover, we let

$$\mathbb{D}_\Gamma := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_{\Gamma, k}$$

be the system of dyadic cubes over a fixed chamber Γ , whose construction will be given in Corollary 3.7.

Lemma 2.7. *There exist 2^N systems of dyadic cubes $\mathbb{D}^1, \mathbb{D}^2, \dots, \mathbb{D}^{2^N}$ such that for any ball $B = B(x_B, r_B) \subset \mathbb{R}^N$ with $\frac{2^{-k-2}}{3} \leq r_B < \frac{2^{-k-1}}{3}$ for some $k \in \mathbb{Z}$, there exists a cube $Q \in \bigcup_{v=1}^{2^N} \mathbb{D}_k^v$ such that*

$$B \subset Q \subset 12 \sqrt{N} B.$$

Proof. • **Step 1.** Recall the one-dimensional setting given in [10, Section 2].

Let $\mathbb{D}^{(0)}(\mathbb{R})$ be the standard system of dyadic intervals over $\mathbb{R}$ (see Notation 2.6). Let $\delta = 1/3$ and define

$$\mathbb{D}^{(\delta)}(\mathbb{R}) = \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k^{(\delta)}(\mathbb{R}),$$

where for $k \geq 0$,

$$\mathbb{D}_k^{(\delta)}(\mathbb{R}) = \{I + \delta \mid I \in \mathbb{D}_k^{(0)}(\mathbb{R})\},$$

while for $k < 0$ and k even,

$$\mathbb{D}_k^{(\delta)}(\mathbb{R}) = \left\{ \left\lfloor \frac{m}{2^k} + \delta + \sum_{j=(k/2)+1}^0 2^{-2j} \right\rfloor, \frac{m+1}{2^k} + \delta + \sum_{j=(k/2)+1}^0 2^{-2j} \right\} \mid m \in \mathbb{Z}.$$

These choices together with the nestedness property completely determine the collections $\mathbb{D}_k^{(\delta)}(\mathbb{R})$ for $k < 0$, k odd.

Fix an interval $J \subset \mathbb{R}$ with $2^{-k-1}/3 \leq |J| < 2^{-k}/3$ for some $k \in \mathbb{Z}$. Now we set $A_k = \{m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for every fixed k and

- (1) $A_k^{(\delta)} = \{\delta + m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for $k \geq 0$,
- (2) $A_k^{(\delta)} = \{\delta + \sum_{j=(k/2)+1}^0 2^{-2j} + m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for $k < 0$, k even, and
- (3) $A_k^{(\delta)} = \{\delta + \sum_{j=(k-1)/2+1}^0 2^{-2j} + m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for $k < 0$, k odd.

Note that for any two different points $a, b \in A_k \cup A_k^{(\delta)}$, we have $|a - b| \geq 2^{-k}/3 > |J|$. Thus, there is at most one element of $A_k \cup A_k^{(\delta)}$ contained in J . Hence, $J \cap A_k = \emptyset$ or $J \cap A_k^{(\delta)} = \emptyset$. Therefore, J must be contained in some dyadic interval $I \in \mathbb{D}_k^{(0)}$ or $I \in \mathbb{D}_k^{(\delta)}$ and $|I| = 2^{-k} \leq 6|J|$.

• **Step 2.** We extend the construction to $\mathbb{R}^N$ ($N \geq 1$).

For $\vartheta = (\vartheta_1, \vartheta_2, \dots, \vartheta_N) \in \{0, \delta\}^N$, define

$$\mathbb{D}^{(\vartheta)} := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k^{(\vartheta)}, \quad \mathbb{D}_k^{(\vartheta)} := \left\{ \prod_{j=1}^N I_j : I_j \in \mathbb{D}_k^{(\vartheta_j)} \right\}.$$

Applying Step 1 to $\pi_i(B)$, where $\pi_i(B)$ denotes the i -th coordinate projection, we obtain $I_i \in \mathbb{D}_k^{(0)}$ or $I_i \in \mathbb{D}_k^{(\delta)}$ with $\pi_i(B) \subset I_i$ and $|I_i| = 2^{-k} \leq 12r_B$. Let $\vartheta_i \in \{0, \delta\}$ be the corresponding shift so that $I_i \in \mathbb{D}_k^{(\vartheta_i)}$, set $\vartheta := (\vartheta_1, \dots, \vartheta_N)$, and let $Q = \prod_{j=1}^N I_j \in \mathbb{D}_k^{(\vartheta)}$. It follows that $B \subset Q$ and $\ell(Q) = 2^{-k} \leq 12r_B$. Hence, for every $y \in Q$,

$$\|y - x_B\| \leq \sqrt{N} \ell(Q) \leq 12 \sqrt{N} r_B.$$

This implies that $Q \subset B(x_B, 12 \sqrt{N} r_B)$. Enumerating $\{\mathbb{D}^{(\vartheta)} : \vartheta \in \{0, \delta\}^N\}$ as $\mathbb{D}^1, \dots, \mathbb{D}^{2^N}$, we complete the proof of Lemma 2.7. $\square$

2.3. Haar Basis. Given a system of dyadic cubes $\mathcal{D}$ over $\mathbb{R}^N$, we now recall the standard way to construct an orthonormal Haar system corresponding to $\mathcal{D}$. For each $k \in \mathbb{Z}$ and $Q \in \mathcal{D}_k$, denote by $Ch(Q) := \{R \in \mathcal{D}_{k+1} : R \subseteq Q\}$. Moreover, we let

$$h_Q^0 := w(Q)^{-1/2} \chi_Q.$$

Fix an enumeration $\{Q_i\}_{i=1}^{2^N}$ of $Ch(Q)$. Let

$$\hat{Q}_i := \bigcup_{j=i}^{2^N} Q_j$$

and

$$h_Q^i := \sqrt{\frac{w(Q_i)w(\hat{Q}_{i+1})}{w(\hat{Q}_i)}} \left(\frac{\chi_{Q_i}}{w(Q_i)} - \frac{\chi_{\hat{Q}_{i+1}}}{w(\hat{Q}_{i+1})} \right), \quad i = 1, 2, \dots, 2^N - 1.$$

Lemma 2.8. *For any $1 < p < \infty$ and $f \in L^p(\mathbb{R}^N, dw)$, we have*

$$f(x) = \sum_{Q \in \mathcal{D}} \sum_{i=1}^{2^N-1} \langle f, h_Q^i \rangle_{L^2(\mathbb{R}^N, dw)} h_Q^i(x),$$

where the sum converges both in the $L^p(\mathbb{R}^N, dw)$ norm and pointwise almost everywhere.

Lemma 2.9. *The family of Haar functions $\{h_Q^i\}_{Q \in \mathcal{D}, 0 \leq i \leq 2^N-1}$ has the following properties:*

- (i) h_Q^i is supported on Q for $i \in \{0, 1, \dots, 2^N - 1\}$;
- (ii) h_Q^i is constant on each $R \in Ch(Q)$ for $i \in \{0, 1, \dots, 2^N - 1\}$;
- (iii) $\int_{\mathbb{R}^N} h_Q^i(x) dw(x) = 0$, for $i \in \{1, 2, \dots, 2^N - 1\}$;
- (iv) $\langle h_Q^i, h_Q^j \rangle_{L^2(\mathbb{R}^N, dw)} = 0$ for $i \neq j$, $i, j \in \{1, 2, \dots, 2^N - 1\}$;
- (v) The collection $\{h_Q^i\}_{i=0}^{2^N-1}$ is an orthogonal basis for the vector space $V(Q)$ of all functions on Q that are constant on each sub-cube $R \in Ch(Q)$.
- (vi) $\|h_Q^i\|_{L^p(\mathbb{R}^N, dw)} \approx w(Q)^{\frac{1}{p}-\frac{1}{2}}$ for $1 \leq p \leq \infty$ and $i \in \{1, 2, \dots, 2^N - 1\}$;

2.4. Nearly weakly orthonormal sequences. Rochberg and Semmes [12] introduced the notion of *nearly weakly orthonormal (NWO)* sequences of functions. This notion has since proved highly effective in characterizing Schatten class membership of singular integral commutators in a variety of contexts (see [2–4, 8, 9, 11–14]). The properties of NWO sequences were originally investigated in the Euclidean setting (equipped with Lebesgue measure), but the extension to spaces of homogeneous type is straightforward once a system of dyadic cubes is available in this general framework (see [8, Section 13] for a complete proof). For our purposes, we adapt the extension to the Dunkl setting $(\mathbb{R}^N, \|\cdot\|, w)$ rather than recalling the general definition of NWO sequences and the general extension to spaces of homogeneous type.

Let $\mathcal{D}$ be a system of dyadic cubes. A sequence of functions $\{e_Q\}_{Q \in \mathcal{D}}$ is an NWO sequence if

- (1) $\text{supp } e_Q \subseteq cQ$ for some positive constant c independent of Q
- (2) $\|e_Q\|_{L^s(\mathbb{R}^N, dw)} \leq w(Q)^{1/s-1/2}$ for some $s > 2$.

Lemma 2.10. *Let $1 < p < \infty$, and let $\mathcal{D}$ be a system of dyadic cubes over $\mathbb{R}^N$. For any compact operator T on $L^2(\mathbb{R}^N, dw)$, we have*

$$\left(\sum_{Q \in \mathcal{D}} |\langle T e_Q, f_Q \rangle_{L^2(\mathbb{R}^N, dw)}|^p \right)^{1/p} \leq C_{N,p,w} \|T\|_{S^p},$$

where $\{e_Q\}_{Q \in \mathcal{D}}$ and $\{f_Q\}_{Q \in \mathcal{D}}$ are NWO sequences.

Proof. This property can be found in [12, (1.10)] (see also [8, Section 7]). $\square$

Lemma 2.11. *Let $0 < p < \infty$, $0 < q \leq \infty$, and let $\mathcal{D}$ be a system of dyadic cubes over $\mathbb{R}^N$. Assume that*

$$A = \sum_{Q \in \mathcal{D}} \lambda_Q \langle \cdot, e_Q \rangle_{L^2(\mathbb{R}^N, dw)} f_Q,$$

where $\{e_Q\}_{Q \in \mathcal{D}}$ and $\{f_Q\}_{Q \in \mathcal{D}}$ are NWO sequences and $\{\lambda_Q\}_{Q \in \mathcal{D}}$ is a sequence of scalars. We have

$$(2.5) \quad \|A\|_{S^{p,q}} \leq C_{N,p,q,w} \|\{\lambda_Q\}_Q\|_{\ell^{p,q}}.$$

Proof. This property can be found in [12, (1.9)] (see also [8, Section 7]). $\square$

3. ANALYTIC TOOLS FOR WEYL CHAMBER GEOMETRY

Convention. Throughout this paper, we will use the following notation:

- Let $R \subset \mathbb{R}^N$ be a finite normalized reduced root system;
- Let $W := \text{span}(R)$ with $N_0 := \dim W$;
- Let $\Omega := \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}$;
- Let Γ_+ be a fixed open Weyl chamber (see Definition 3.1) and Γ be its closure.

Definition 3.1. [6, Definition 2.7] *A connected component of*

$$(3.1) \quad \{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \text{ for all } v \in R\}$$

is called an open Weyl chamber. Its closure is called a Weyl chamber. Define the corresponding sets of positive and negative roots by

$$R_+ := \{\alpha \in R : \langle x, \alpha \rangle > 0 \text{ for all } x \in \Gamma_+\}, \quad R_- := -R_+.$$

Definition 3.2. [6, Definition 2.9] *A root $\alpha \in R_+$ is called simple in R_+ if it cannot be written as*

$$\alpha = \lambda_1 \alpha_1 + \lambda_2 \alpha_2 \quad \text{with } \lambda_1, \lambda_2 \geq 1 \text{ and } \alpha_1, \alpha_2 \in R_+.$$

The set $\Delta \subset R_+$ of all simple roots is called the simple system.

Lemma 3.3. *With the notation given in Definition 3.2, Δ is a basis of W . If we denote by $\Delta = \{\alpha_1, \alpha_2, \dots, \alpha_{N_0}\}$, then the open and closed chambers can be written as*

$$(3.2) \quad \Gamma_+ = \{x \in \mathbb{R}^N : \langle x, \alpha_i \rangle > 0, i = 1, \dots, N_0\},$$

$$(3.3) \quad \Gamma = \{x \in \mathbb{R}^N : \langle x, \alpha_i \rangle \geq 0, i = 1, \dots, N_0\},$$

and every root $\alpha \in R$ admits a unique expansion

$$\alpha = \sum_{i=1}^{N_0} c_i \alpha_i$$

such that either $c_i \geq 0$ for all i (when $\alpha \in R_+$), or $c_i \leq 0$ for all i (when $\alpha \in R_-$). In particular, the simple system Δ is uniquely determined by Γ_+ (equivalently, by Γ).

Proof. Let $\Delta = \{\alpha_1, \dots, \alpha_n\} \subset R_+$ be the set of simple roots in R_+ . By [6, Proposition 2.10], every $\alpha \in R_+$ is a nonnegative linear combination of $\alpha_1, \dots, \alpha_n$. Since $R = R_+ \cup R_-$ (this follows from the fact that for each $\alpha \in R$, the continuous function $x \mapsto \langle x, \alpha \rangle$ does not vanish on the connected set Γ_+) and $R_- = -R_+$, it follows that $R \subset \text{span}(\Delta)$ and hence,

$$W = \text{span}(R) \subset \text{span}(\Delta).$$

The reverse inclusion $\text{span}(\Delta) \subset W$ is immediate from $\Delta \subset R \subset W$, so $\text{span}(\Delta) = W$. On the other hand, it follows from [6, Theorem 2.16] that the simple roots are linearly independent. Consequently, Δ is a basis of W , and therefore $n = \dim W = N_0$. Moreover, every $\alpha \in R$ has a unique expansion $\alpha = \sum_{i=1}^{N_0} c_i \alpha_i$. If $\alpha \in R_+$, the coefficients can be chosen nonnegative by [6, Proposition 2.10]. If $\alpha \in R_- = -R_+$, then $-\alpha \in R_+$ has an expansion with all coefficients nonnegative, and multiplying by -1 gives an expansion of α with all coefficients nonpositive. The descriptions (3.2) and (3.3) follow from [6, Corollary 2.11].

To see that Δ is uniquely determined by Γ_+ , note that Γ_+ determines R_+ by Definition 3.1, and the set of simple roots in R_+ is characterized intrinsically by Definition 3.2. Finally, since Γ_+ is the (unique) interior component of Γ , specifying Γ determines Γ_+ and vice versa. This completes the proof of Lemma 3.3. $\square$

Notation 3.4. With the notation given in Lemma 3.3, denote by

$$\Delta = \{\alpha_1, \dots, \alpha_{N_0}\}$$

the simple system associated with Γ .

Lemma 3.5. Each Weyl chamber Γ admits the decomposition

$$\Gamma = K(\Gamma) \times W^\perp, \quad K(\Gamma) := \{w \in W : \langle w, \alpha \rangle \geq 0 \text{ for all } \alpha \in \Delta\}.$$

Proof. Write $x = w + z$ with $w \in W$ and $z \in W^\perp$. Since $\alpha_i \in W$, we have $\langle x, \alpha_i \rangle = \langle w, \alpha_i \rangle$ for all $i = 1, \dots, N_0$. This, together with (3.3), implies that $x \in \Gamma$ if and only if $w \in K(\Gamma)$. This implies that $\Gamma = K(\Gamma) \times W^\perp$, and thus completes the proof. $\square$

Corollary 3.6. Let $\Phi_{W^\perp}$ be any linear isomorphism from $W^\perp$ to $\mathbb{R}^{N-N_0}$, and let

$$\begin{aligned} \Phi_W : W &\longrightarrow \mathbb{R}^{N_0}, & \Phi_W(w) &:= (\langle w, \alpha_1 \rangle, \dots, \langle w, \alpha_{N_0} \rangle), \\ \Phi : \mathbb{R}^N = W \oplus W^\perp &\longrightarrow \mathbb{R}^{N_0} \oplus \mathbb{R}^{N-N_0} \cong \mathbb{R}^N, & \Phi(w + z) &:= (\Phi_W(w), \Phi_{W^\perp}(z)). \end{aligned}$$

Then Φ is a linear isomorphism and

$$\Phi(\Gamma) = \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}.$$

Proof. By Lemma 3.5 we have $\Gamma = K(\Gamma) \times W^\perp$, where

$$K(\Gamma) = \{w \in W : \langle w, \alpha_i \rangle \geq 0, i = 1, \dots, N_0\}.$$

Since Δ is a basis of W , the Gram matrix $G = (\langle \alpha_i, \alpha_j \rangle)_{i,j=1}^{N_0}$ is nonsingular. Hence, $\Phi_W : W \rightarrow \mathbb{R}^{N_0}$ is a linear isomorphism. Moreover, for $w \in W$ we have $w \in K(\Gamma)$ if and only if $\Phi_W(w) \in \mathbb{R}_+^{N_0}$, so

$$\Phi_W(K(\Gamma)) = \mathbb{R}_+^{N_0}.$$

By construction, Φ is a linear isomorphism (as a direct sum of isomorphisms), and therefore

$$\Phi(\Gamma) = \Phi(K(\Gamma) \times W^\perp) = \Phi_W(K(\Gamma)) \times \Phi_{W^\perp}(W^\perp) = \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0},$$

which proves the desired conclusion. $\square$

Corollary 3.7. *Let Γ be an arbitrary Weyl chamber. There exists a system of dyadic cubes over Γ .*

Proof. Let $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ be the linear isomorphism from Corollary 3.6 so that

$$\Phi(\Gamma) = \Omega := \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}.$$

For each $k \in \mathbb{Z}$, define $\mathbb{D}_{\Omega,k}^0$ to be the family of half-open cubes

$$Q = \prod_{i=1}^N I_i, \quad I_i = [m_i 2^{-k}, (m_i + 1) 2^{-k}),$$

where $m_i \in \mathbb{Z}_+$ for $1 \leq i \leq N_0$ and $m_i \in \mathbb{Z}$ for $N_0 < i \leq N$. Set

$$\mathbb{D}_\Omega^0 := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_{\Omega,k}^0.$$

It is standard that $\mathbb{D}_\Omega^0$ is a system of dyadic cubes over Ω .

Next, for each $k \in \mathbb{Z}$, define

$$\mathbb{D}_{\Gamma,k} := \{\Phi^{-1}(Q) : Q \in \mathbb{D}_{\Omega,k}^0\}, \quad \mathbb{D}_\Gamma := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_{\Gamma,k}.$$

Since $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ is a linear isomorphism, we see that $\mathbb{D}_\Gamma$ is a system of dyadic cubes over Γ . $\square$

Notation 3.8. *Let Φ be the linear isomorphism given in Corollary 3.6. For every Borel set $E \subset \Omega$, define*

$$(3.4) \quad \mu(E) := w(\Phi^{-1}(E)).$$

Lemma 3.9. *We have*

$$(3.5) \quad d\mu(u, z) = \phi(u) du dz, \quad \phi(u) = |\det(\Phi)|^{-1} \prod_{v \in R_+} \left(\sum_{i=1}^{N_0} a_{v,i} u_i \right)^{2k(v)}.$$

where for each $v \in R_+$, the coefficients $a_{v,i} \geq 0$ are uniquely determined by the simple-root expansion $v = \sum_{i=1}^{N_0} a_{v,i} \alpha_i$ and satisfy $(a_{v,1}, \dots, a_{v,N_0}) \neq \mathbf{0}$.

Proof. Since $\Phi(\Gamma) = \Omega$, we have $\Phi^{-1}(\Omega) = \Gamma$. Hence, for every nonnegative Borel function f on Ω ,

$$\int_{\Omega} f(u, z) d\mu(u, z) = \int_{\Gamma} f(\Phi(x)) dw(x).$$

We make the linear change of variables $(u, z) = \Phi(x)$, so that $x = \Phi^{-1}(u, z)$. Since Φ is linear and invertible, the Jacobian is constant and

$$dx = |\det(\Phi)|^{-1} du dz.$$

It follows from the definition of Φ that $u_i = \langle x, \alpha_i \rangle$ for $i = 1, \dots, N_0$. This implies that

$$(3.6) \quad \langle x, v \rangle = \left\langle x, \sum_{i=1}^{N_0} a_{v,i} \alpha_i \right\rangle = \sum_{i=1}^{N_0} a_{v,i} \langle x, \alpha_i \rangle = \sum_{i=1}^{N_0} a_{v,i} u_i.$$

Substituting these identities into the previous integral, we obtain

$$\begin{aligned} \int_{\Omega} f(u, z) d\mu(u, z) &= \int_{\Gamma} f(\Phi(x)) \left(\prod_{v \in R_+} |\langle x, v \rangle|^{2k(v)} \right) dx \\ &= \int_{\Omega} f(u, z) |\det(\Phi)|^{-1} \prod_{v \in R_+} \left(\sum_{i=1}^{N_0} a_{v,i} u_i \right)^{2k(v)} du dz. \end{aligned}$$

Therefore, (3.5) holds. This completes the proof of Lemma 3.9. $\square$

Lemma 3.10. *Let Γ_1 and Γ_2 be two Weyl chambers. Then there exist an integer $M \geq 0$ and $\alpha_1, \dots, \alpha_M \in R$ such that, if we set $\Gamma^{(0)} := \Gamma_2$ and*

$$\Gamma^{(k)} := \sigma_{\alpha_k}(\Gamma^{(k-1)}), \quad k = 1, \dots, M,$$

then the following hold:

- (1) $\alpha_k \in \Delta(\Gamma^{(k-1)})$ for each $k = 1, \dots, M$;
- (2) $\Gamma^{(M)} = \Gamma_1$.

Proof. If $\Gamma_1 = \Gamma_2$, we take $M = 0$ and there is nothing to prove. Assume $\Gamma_1 \neq \Gamma_2$. Since the Weyl group G acts transitively on the set of Weyl chambers (see [6, Theorem 2.12 (2)]), there exists $g \in G$ such that $\Gamma_1 = g(\Gamma_2)$.

Let $\Delta(\Gamma_2) = \{\beta_1, \beta_2, \dots, \beta_{N_0}\}$ be the simple system associated with Γ_2 . By [6, Theorem 2.12(4)], the Weyl group G is generated by the simple reflections $\sigma_{\beta_1}, \sigma_{\beta_2}, \dots, \sigma_{\beta_{N_0}}$. Hence, there exist indices $i_1, \dots, i_M \in \{1, \dots, N_0\}$ such that

$$g = \sigma_{\beta_{i_1}} \sigma_{\beta_{i_2}} \cdots \sigma_{\beta_{i_M}}.$$

Define

$$w_0 := \text{Id}, \quad w_k := \sigma_{\beta_{i_1}} \cdots \sigma_{\beta_{i_k}} \quad (k = 1, \dots, M),$$

and set

$$\Gamma^{(k)} := w_k(\Gamma_2) \quad (k = 0, 1, \dots, M).$$

It follows that $\Gamma^{(0)} = \Gamma_2$ and $\Gamma^{(M)} = w_M(\Gamma_2) = g(\Gamma_2) = \Gamma_1$, which implies the statement (2).

For each $k = 1, \dots, M$, define

$$\alpha_k := w_{k-1}(\beta_{i_k}) \in R.$$

The rest of the proof is divided into two steps.

• **Step 1.** Verify that $\Gamma^{(k)} = \sigma_{\alpha_k}(\Gamma^{(k-1)})$.

Since each $w \in G$ is orthogonal, for any root $\beta \in R$ one has the conjugation identity

$$w \sigma_{\beta} w^{-1} = \sigma_{w(\beta)}, \quad w \in G.$$

Applying this with $w = w_{k-1}$ and $\beta = \beta_{i_k}$ yields

$$\sigma_{\alpha_k} = \sigma_{w_{k-1}(\beta_{i_k})} = w_{k-1} \sigma_{\beta_{i_k}} w_{k-1}^{-1}.$$

Therefore,

$$\sigma_{\alpha_k}(\Gamma^{(k-1)}) = w_{k-1} \sigma_{\beta_{i_k}} w_{k-1}^{-1}(w_{k-1}(\Gamma_2)) = w_{k-1} \sigma_{\beta_{i_k}}(\Gamma_2) = w_k(\Gamma_2) = \Gamma^{(k)}.$$

• **Step 2.** Verify that $\alpha_k \in \Delta(\Gamma^{(k-1)})$.

Recall that if Γ is a Weyl chamber associated with simple system $\Delta(\Gamma)$, then

$$\Gamma = \{x \in \mathbb{R}^N : \langle x, \gamma \rangle \geq 0 \text{ for all } \gamma \in \Delta(\Gamma)\}.$$

Applying $w_{k-1} \in G$ to the above description for Γ_2 and using $\langle w_{k-1}x, \beta \rangle = \langle x, w_{k-1}^{-1}\beta \rangle$, we obtain

$$\Gamma^{(k-1)} = w_{k-1}(\Gamma_2) = \{x \in \mathbb{R}^N : \langle x, w_{k-1}(\beta_j) \rangle \geq 0 \text{ for } j = 1, \dots, N_0\}.$$

By the uniqueness of the simple system associated with a Weyl chamber (see Lemma 3.3), we conclude that

$$\Delta(\Gamma^{(k-1)}) = \Delta(w_{k-1}(\Gamma_2)) = w_{k-1}(\Delta(\Gamma_2)) = \{w_{k-1}(\beta_1), \dots, w_{k-1}(\beta_{N_0})\}.$$

In particular, $\alpha_k = w_{k-1}(\beta_{i_k}) \in \Delta(\Gamma^{(k-1)})$, proving the first statement (1).

This completes the proof of Lemma 3.10. □

Definition 3.11. We say that $R' \subset \Omega$ is an axis-parallel rectangle if

$$R' := I_1 \times \dots \times I_N,$$

with $I_i = [I_i^-, I_i^+] \subset [0, \infty)$ for $1 \leq i \leq N_0$ and $I_i = [I_i^-, I_i^+] \subset \mathbb{R}$ for $i > N_0$.

Denote by

$$B_{\Omega}(x_0, r) := \{x \in \Omega : \|x - x_0\| < r\}.$$

The following lemma demonstrates that $(\Omega, \|\cdot\|, d\mu)$ is a doubling metric measure space.

Lemma 3.12. There exists a constant $C \geq 1$ such that for every $x_0 = (u_0, z_0) \in \Omega$ and every $r > 0$,

$$(3.7) \quad C^{-1} r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} \leq \mu(B_{\Omega}(x_0, r)) \leq C r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)},$$

where

$$L_v(u) := \sum_{i=1}^{N_0} a_{v,i} u_i, \quad L_{v,0} := L_v(u_0) = \sum_{i=1}^{N_0} a_{v,i} (u_0)_i.$$

In particular, there exists a constant $C_{\text{dbl}, \Omega} \geq 1$ such that for every $x \in \Omega$ and every $r > 0$,

$$\mu(B_{\Omega}(x, 2r)) \leq C_{\text{dbl}, \Omega} \mu(B_{\Omega}(x, r)).$$

Proof. Fix $x_0 = (u_0, z_0) \in \Omega$ and $r > 0$. For each $v \in R_+$ define the linear form

$$L_v(u) := a_v \cdot u, \quad a_v := (a_{v,1}, \dots, a_{v,N_0}) \in \mathbb{R}_+^{N_0} \setminus \{0\}.$$

Since R_+ is finite and each $a_v \neq 0$, we have

$$L_* := \max_{v \in R_+} \|a_v\| < \infty, \quad m_* := \min_{v \in R_+} \sum_{i=1}^{N_0} a_{v,i} > 0.$$

Upper bound. Let $x = (u, z) \in B_\Omega(x_0, r)$. Then

$$\|u - u_0\| \leq \|x - x_0\| \leq r.$$

Hence, for every $v \in R_+$,

$$|L_v(u) - L_{v,0}| = |a_v \cdot (u - u_0)| \leq \|a_v\| \cdot \|u - u_0\| \leq L_* r.$$

Since $L_v(u) \geq 0$ on Ω (all coefficients and coordinates are nonnegative), we obtain

$$L_v(u) \leq L_{v,0} + L_* r \leq C_1 (L_{v,0} + r), \quad C_1 := \max\{1, L_*\}.$$

Therefore,

$$\phi(u) = \prod_{v \in R_+} L_v(u)^{2k(v)} \leq C_1^{2 \sum_{v \in R_+} k(v)} \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} =: C_2 \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)}.$$

Integrating over $B_\Omega(x_0, r)$ and using $|B_\Omega(x_0, r)| \leq |B(x_0, r)| = \omega_N r^N$, where ω_N is the volume of the unit ball in $\mathbb{R}^N$, gives

$$\mu(B_\Omega(x_0, r)) = \int_{B_\Omega(x_0, r)} \phi(u) dx \leq C_2 |B(x_0, r)| \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} \leq C_3 r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)},$$

with $C_3 := C_2 \omega_N$.

Lower bound. We construct a smaller Euclidean ball B_c contained in $B_\Omega(x_0, r)$ on which ϕ admits a uniform positive lower bound in terms of $\prod_{v \in R_+} (L_{v,0} + r)^{2k(v)}$. To this end, we first choose

$$\lambda := 1 + \frac{L_*}{m_*} > 1, \quad \eta := \frac{1}{4(\sqrt{N_0} \lambda + 1)}, \quad \theta := \lambda \eta.$$

These parameters depend only on L_* and m_* , and satisfy

$$\theta > \eta > 0, \quad \sqrt{N_0} \theta + \eta = \eta(\sqrt{N_0} \lambda + 1) = \frac{1}{4} < 1.$$

Let $\mathbf{1}_{N_0} := (1, \dots, 1) \in \mathbb{R}^{N_0}$ and define the center

$$x_c := (u_c, z_c) := (u_0 + \theta r \mathbf{1}_{N_0}, z_0), \quad B_c := B(x_c, \eta r).$$

We *claim* that $B_c \subset B_\Omega(x_0, r)$. To see this inclusion, we first apply the triangle inequality to see that for any $x \in B_c$,

$$\|x - x_0\| \leq \|x - x_c\| + \|x_c - x_0\| \leq \eta r + \theta r \sqrt{N_0} \leq r,$$

since $\sqrt{N_0} \theta + \eta < 1$. Hence $B_c \subset B(x_0, r)$. Moreover, for any $x = (u, z) \in B_c$, we have $\|u - u_c\| \leq \|x - x_c\| \leq \eta r$. This implies that for each $1 \leq i \leq N_0$,

$$u_i \geq (u_c)_i - \eta r = (u_0)_i + (\theta - \eta)r \geq 0,$$

since $u_0 \in \mathbb{R}_+^{N_0}$ and $\theta \geq \eta$. Thus $u \in \mathbb{R}_+^{N_0}$ and $x \in \Omega$. Therefore, $B_c \subset B_\Omega(x_0, r)$. This finishes the proof of the claim.

Next we obtain a uniform lower bound for $L_v(u)$ on B_c . Write any $x = (u, z) \in B_c$ as

$$u = u_0 + \theta r \mathbf{1}_{N_0} + \Delta_u, \quad z = z_0 + \Delta_z, \quad \|\Delta_u\|^2 + \|\Delta_z\|^2 \leq (\eta r)^2,$$

so in particular $\|\Delta_u\| \leq \eta r$. Then for all $x = (u, z) \in B_c$ and all $v \in R_+$,

$$\begin{aligned} L_v(u) &= a_v \cdot u = a_v \cdot u_0 + \theta r a_v \cdot \mathbf{1}_{N_0} + a_v \cdot \Delta_u \\ &\geq L_{v,0} + \theta r \sum_{i=1}^{N_0} a_{v,i} - \|a_v\| \|\Delta_u\| \geq L_{v,0} + r K_0 \geq c_0 (L_{v,0} + r). \end{aligned}$$

with

$$K_0 := \theta m_* - \eta L_* = \eta(\lambda m_* - L_*) = \eta m_* > 0, \quad c_0 := \min\{1, K_0\} \in (0, 1].$$

Consequently, for all $u \in B_c$,

$$\phi(u) = \prod_{v \in R_+} L_v(u)^{2k(v)} \geq c_0^{2 \sum_{v \in R_+} k(v)} \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} =: C_4 \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)}.$$

This, in combination with the fact that $B_c \subset B_\Omega(x_0, r)$, yields

$$\mu(B_\Omega(x_0, r)) \geq \mu(B_c) = \int_{B_c} \phi(u) dx \geq C_4 |B_c| \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} = C_5 r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)},$$

where $C_5 := C_4 \omega_N \eta^N > 0$.

The second statement follows immediately from the first one. Indeed, by the first statement,

$$\mu(B_\Omega(x_0, 2r)) \leq C_3 (2r)^N \prod_{v \in R_+} (L_{v,0} + 2r)^{2k(v)} \leq C_3 2^{N+\mathcal{K}} r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} \leq C_3 C_5^{-1} 2^{N+\mathcal{K}} \mu(B_\Omega(x_0, r)),$$

where we set $\mathcal{K} := 2 \sum_{v \in R_+} k(v)$. Therefore, $(\Omega, \|\cdot\|, \mu)$ is doubling with the doubling constant $C_{\text{dbl}, \Omega} := C_3 C_5^{-1} 2^{N+\mathcal{K}}$. This finishes the proof of Lemma 3.12. $\square$

Let $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ be the linear isomorphism given in Corollary 3.6 such that $\Phi(\Gamma) = \Omega$. Since Φ is linear and invertible, there exist positive constants c_Φ and C_Φ such that for any $x, y \in \mathbb{R}^N$,

$$(3.8) \quad c_\Phi \|x - y\| \leq \|\Phi(x) - \Phi(y)\| \leq C_\Phi \|x - y\|.$$

Denote by $B_\Gamma(x, r) := \{y \in \Gamma : \|y - x\| < r\}$. It follows from (3.8) that for any $x_0 \in \Gamma$ and $r > 0$,

$$(3.9) \quad B_\Omega(\Phi(x_0), c_\Phi r) \subset \Phi(B_\Gamma(x_0, r)) \subset B_\Omega(\Phi(x_0), C_\Phi r).$$

This, together with Lemma 3.12, yields

$$(3.10) \quad \mu(\Phi(B_\Gamma(x_0, r))) \simeq \mu(B_\Omega(\Phi(x_0), C_\Phi r)).$$

Lemma 3.13. *For all $x_0 \in \Gamma$ and $r > 0$, one can find an axis-parallel rectangle R' on Ω such that*

$$(3.11) \quad B_\Omega(\Phi(x_0), C_\Phi r) \subset R' \subset B_\Omega(\Phi(x_0), \sqrt{N} C_\Phi r).$$

Furthermore, we have

$$(3.12) \quad \mu(R') \simeq \mu(B_\Omega(\Phi(x_0), C_\Phi r)).$$

Proof. We write $(u_0, z_0) := \Phi(x_0) \in \Omega$. and define the axis-parallel rectangle

$$(3.13) \quad R' := \left(\prod_{i=1}^{N_0} [\max\{0, (u_0)_i - C_\Phi r\}, (u_0)_i + C_\Phi r] \right) \times \left(\prod_{j=1}^{N-N_0} [(z_0)_j - C_\Phi r, (z_0)_j + C_\Phi r] \right).$$

On the one hand, it follows immediately that $B_\Omega(\Phi(x_0), C_\Phi r) \subset R'$. On the other hand, since R' is a rectangle with half-side lengths bounded by $C_\Phi r$ in each coordinate, we have

$$R' \subset B_\Omega(\Phi(x_0), \sqrt{N} C_\Phi r).$$

The second statement follows from the first one together with Lemma 3.12. The proof of Lemma 3.13 is finished. $\square$

The following theorem demonstrates that $(\Gamma, \|\cdot\|, dw)$ is a doubling metric measure space.

Theorem 3.14. *There exists a constant $C \geq 1$ such that for every $x \in \Gamma$ and every $r > 0$,*

$$(3.14) \quad C^{-1} r^N \prod_{v \in R_+} (|\langle x, v \rangle| + r)^{2k(v)} \leq w(B_\Gamma(x, r)) \leq C r^N \prod_{v \in R_+} (|\langle x, v \rangle| + r)^{2k(v)}.$$

In particular, there exists a constant $C_{\text{dbl}, \Gamma} \geq 1$ such that for every $x \in \Gamma$ and every $r > 0$,

$$w(B_\Gamma(x, 2r)) \leq C_{\text{dbl}, \Gamma} w(B_\Gamma(x, r)).$$

Moreover, for all $x \in \Gamma$ and $r > 0$, we have

$$(3.15) \quad w(B_\Gamma(x, r)) \sim w(B(x, r)).$$

Proof. The main strategy of the proof is to transfer the ball estimates from $(\Omega, \|\cdot\|, \mu)$ back to $(\Gamma, \|\cdot\|, dw)$. Fix $x \in \Gamma$ and $r > 0$. It follows from the equality (3.4) that

$$(3.16) \quad w(B_\Gamma(x, r)) = \mu(\Phi(B_\Gamma(x, r))).$$

Applying μ to the inclusions (3.9) and using (3.16) yields

$$(3.17) \quad \mu(B_\Omega(\Phi(x), c_\Phi r)) \leq w(B_\Gamma(x, r)) \leq \mu(B_\Omega(\Phi(x), C_\Phi r)).$$

Write $\Phi(x) = (u, z) \in \Omega$. For each $v \in R_+$ recall the linear form $L_{v,0} := \sum_{i=1}^{N_0} a_{v,i} u_i$. Combining the inequality (3.17) with Lemma 3.12 and the equality (3.6), we obtain that

$$(3.18) \quad w(B_\Gamma(x, r)) \simeq r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} = r^N \prod_{v \in R_+} (|\langle x, v \rangle| + r)^{2k(v)}.$$

This proves (3.14). The second statement follows immediately from (3.14). The third statement follows from the inequalities (3.14) and (2.1). This completes the proof of Theorem 3.14. $\square$

4. CWIKEL'S ESTIMATE IN $N > 2$

Notation 4.1. *Fix a closed Weyl chamber Γ , and let*

$$\Phi : \mathbb{R}^N \longrightarrow \mathbb{R}^N$$

be the linear isomorphism appearing in Corollary 3.6.

Let

$$\mathbb{D}_\Gamma = \bigcup_{j \in \mathbb{Z}} \mathbb{D}_{\Gamma, j}$$

be the dyadic system over Γ constructed in Corollary 3.7. If $I \in \mathbb{D}_{\Gamma,j}$, then

$$I = \Phi^{-1}(Q)$$

for a unique standard dyadic cube $Q \subset \Omega$ with $\ell(Q) = 2^{-j}$, and we put

$$\ell(I) := \ell(Q),$$

set

$$I^* := \Phi^{-1}((2Q) \cap \Omega).$$

Let $K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(x, y)$ denote the integral kernel of $(-\Delta_{\text{Dunkl}})^{-1/2}$.

For $\sigma \in G, I \in \mathbb{D}_{\Gamma}$, and for every $f \in L^r_{\text{loc}}(\mathbb{R}^N, dw)$, define

$$(4.19) \quad \lambda_{\sigma,I}(f) := \ell(I) w(I)^{-1/r} \|f\|_{L^r(\sigma I, dw)}.$$

Lemma 4.2. Assume that $N > 1$. Then for every pair of multi-indices α, β , there exists a constant $C_{\alpha,\beta} > 0$ such that

$$(4.20) \quad \left| \partial_x^\alpha \partial_y^\beta K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(x, y) \right| \leq C_{\alpha,\beta} \frac{d(x, y)^{1-|\alpha|-|\beta|}}{V(x, y, d(x, y))}$$

for all $x, y \in \mathbb{R}^N$ satisfying $d(x, y) > 0$.

Proof. Let $h_t(x, y)$ denote the integral kernel of the Dunkl heat semigroup $e^{t\Delta_{\text{Dunkl}}}$. By [1, Theorem 4.1(d)], for every pair of multi-indices α, β , there exist constants $C_{\alpha,\beta}, c_{\alpha,\beta} > 0$ such that

$$(4.21) \quad \left| \partial_x^\alpha \partial_y^\beta h_t(x, y) \right| \leq C_{\alpha,\beta} t^{-\frac{|\alpha|+|\beta|}{2}} V(x, y, \sqrt{t})^{-1} \exp\left(-c_{\alpha,\beta} \frac{d(x, y)^2}{t}\right)$$

for all $t > 0$ and $x, y \in \mathbb{R}^N$.

By the spectral functional calculus,

$$(-\Delta_{\text{Dunkl}})^{-1/2} = \frac{1}{\sqrt{\pi}} \int_0^\infty e^{t\Delta_{\text{Dunkl}}} \frac{dt}{\sqrt{t}}.$$

Consequently, for $d(x, y) > 0$,

$$(4.22) \quad K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(x, y) = \frac{1}{\sqrt{\pi}} \int_0^\infty h_t(x, y) \frac{dt}{\sqrt{t}}.$$

It follows that $K_{(-\Delta_{\text{Dunkl}})^{-1/2}}$ is smooth whenever $d(x, y) > 0$ and

$$(4.23) \quad \partial_x^\alpha \partial_y^\beta K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(x, y) = \frac{1}{\sqrt{\pi}} \int_0^\infty \partial_x^\alpha \partial_y^\beta h_t(x, y) \frac{dt}{\sqrt{t}}.$$

Set $m := |\alpha| + |\beta|$. Fix $x, y \in \mathbb{R}^N$ with $d(x, y) > 0$. By (4.21) and (4.23),

$$(4.24) \quad \left| \partial_x^\alpha \partial_y^\beta K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(x, y) \right| \lesssim \int_0^\infty t^{-\frac{m+1}{2}} V(x, y, \sqrt{t})^{-1} \exp\left(-c \frac{d(x, y)^2}{t}\right) dt.$$

Suppose first that $0 < t \leq d(x, y)^2$. Then $\sqrt{t} \leq d(x, y)$. By the upper estimate in (2.2), for $z = x, y$,

$$\frac{w(B(z, d(x, y)))}{w(B(z, \sqrt{t}))} \lesssim \left(\frac{d(x, y)}{\sqrt{t}} \right)^N.$$

Equivalently,

$$w(B(z, \sqrt{t})) \gtrsim \left(\frac{\sqrt{t}}{d(x, y)} \right)^N w(B(z, d(x, y))).$$

Taking the maximum over $z = x, y$ gives

$$(4.25) \quad V(x, y, \sqrt{t}) \gtrsim \left(\frac{\sqrt{t}}{d(x, y)} \right)^N V(x, y, d(x, y)).$$

Therefore,

$$\begin{aligned} & \int_0^{d(x, y)^2} t^{-\frac{m+1}{2}} V(x, y, \sqrt{t})^{-1} \exp\left(-c \frac{d(x, y)^2}{t}\right) dt \\ & \lesssim \frac{d(x, y)^N}{V(x, y, d(x, y))} \int_0^{d(x, y)^2} t^{-\frac{N+m+1}{2}} \exp\left(-c \frac{d(x, y)^2}{t}\right) dt. \end{aligned}$$

Using again the change of variables $s = \frac{d(x, y)^2}{t}$, we obtain

$$\int_0^{d(x, y)^2} t^{-\frac{N+m+1}{2}} \exp\left(-c \frac{d(x, y)^2}{t}\right) dt = d(x, y)^{1-N-m} \int_1^\infty s^{\frac{N+m-3}{2}} e^{-cs} ds \lesssim d(x, y)^{1-N-m}.$$

Consequently,

$$(4.26) \quad \int_0^{d(x, y)^2} t^{-\frac{m+1}{2}} V(x, y, \sqrt{t})^{-1} \exp\left(-c \frac{d(x, y)^2}{t}\right) dt \lesssim \frac{d(x, y)^{1-m}}{V(x, y, d(x, y))}.$$

Suppose next that $t \geq d(x, y)^2$. Then $\sqrt{t} \geq d(x, y)$. By the lower estimate in (2.2), for $z = x, y$,

$$\frac{w(B(z, \sqrt{t}))}{w(B(z, d(x, y)))} \gtrsim \left(\frac{\sqrt{t}}{d(x, y)} \right)^N.$$

It follows that

$$(4.27) \quad V(x, y, \sqrt{t}) \gtrsim \left(\frac{\sqrt{t}}{d(x, y)} \right)^N V(x, y, d(x, y)).$$

Hence

$$\int_{d(x, y)^2}^\infty t^{-\frac{m+1}{2}} V(x, y, \sqrt{t})^{-1} \exp\left(-c \frac{d(x, y)^2}{t}\right) dt \lesssim \frac{d(x, y)^N}{V(x, y, d(x, y))} \int_{d(x, y)^2}^\infty t^{-\frac{N+m+1}{2}} dt.$$

Since $N > 1$ and $m \geq 0$, we have $N + m > 1$, and therefore

$$\int_{d(x, y)^2}^\infty t^{-\frac{N+m+1}{2}} dt = \frac{2}{N + m - 1} d(x, y)^{1-N-m}.$$

Thus

$$(4.28) \quad \int_{d(x, y)^2}^\infty t^{-\frac{m+1}{2}} V(x, y, \sqrt{t})^{-1} \exp\left(-c \frac{d(x, y)^2}{t}\right) dt \lesssim \frac{d(x, y)^{1-m}}{V(x, y, d(x, y))}.$$

Combining (4.24), (4.26), and (4.28), we obtain

$$|\partial_x^\alpha \partial_y^\beta K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(x, y)| \lesssim \frac{d(x, y)^{1-m}}{V(x, y, d(x, y))}.$$

Since $m = |\alpha| + |\beta|$, this gives

$$\left| \partial_x^\alpha \partial_y^\beta K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(x, y) \right| \lesssim \frac{d(x, y)^{1-|\alpha|-|\beta|}}{V(x, y, d(x, y))},$$

which is exactly (4.20). $\square$

Lemma 4.3. *Assume that $N > 1$ and $2 < r < \infty$. Then there exist a constant $C > 0$, nonnegative numbers $\{c_{p,q}\}_{p,q \in \mathbb{Z}^N}$, functions $\{e_{p,q,I}^{\sigma,\tau}\}_{p,q \in \mathbb{Z}^N, I \in \mathbb{D}_\Gamma, \sigma, \tau \in G}$, $\{h_{p,q,I}^{\sigma,\tau}\}_{p,q \in \mathbb{Z}^N, I \in \mathbb{D}_\Gamma, \sigma, \tau \in G}$, such that*

(1) *for every $M > 0$,*

$$(4.29) \quad c_{p,q} \leq C_M (1 + |p| + |q|)^{-M}, \quad p, q \in \mathbb{Z}^N;$$

(2)

$$(4.30) \quad \text{supp } e_{p,q,I}^{\sigma,\tau} \subset \sigma I \quad \text{supp } h_{p,q,I}^{\sigma,\tau} \subset \tau I^*;$$

(3)

$$(4.31) \quad \|e_{p,q,I}^{\sigma,\tau}\|_{L^r(\mathbb{R}^N, dw)} \leq w(I)^{\frac{1}{r}-\frac{1}{2}}, \quad \|h_{p,q,I}^{\sigma,\tau}\|_{L^r(\mathbb{R}^N, dw)} \leq w(I)^{\frac{1}{r}-\frac{1}{2}};$$

(4) *For every $f \in L^r_{\text{loc}}(\mathbb{R}^N, dw)$ and for almost every (x, y) with $d(x, y) > 0$,*

$$(4.32) \quad f(x) K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(x, y) = C \sum_{\sigma, \tau \in G} \sum_{p, q \in \mathbb{Z}^N} c_{p,q} \sum_{I \in \mathbb{D}_\Gamma} \lambda_{\sigma, I}(f) e_{p,q,I}^{\sigma,\tau}(x) h_{p,q,I}^{\sigma,\tau}(y).$$

Proof. Fix $f \in L^r_{\text{loc}}(\mathbb{R}^N, dw)$. We divide the proof into several steps.

Step 1.

Fix $\sigma, \tau \in G$, and suppose that $x \in \sigma(\Gamma)^\circ$, $y \in \tau(\Gamma)^\circ$. Set $x_\sigma := \sigma^{-1}x \in \Gamma^\circ$, $y_\tau := \tau^{-1}y \in \Gamma^\circ$, and

$$u := \Phi(x_\sigma), \quad v := \Phi(y_\tau).$$

Since x_σ and y_τ belong to the same closed Weyl chamber, (2.4) gives

$$d(x, y) = d(x_\sigma, y_\tau) = \|x_\sigma - y_\tau\|.$$

Together with (3.8), this yields

$$(4.33) \quad C_\Phi^{-1}|u - v| \leq d(x, y) \leq c_\Phi^{-1}|u - v|.$$

Choose $\rho \in C_c^\infty(\mathbb{R}^N)$ such that

$$0 \leq \rho \leq 1, \quad \rho(z) = 1 \quad \text{for } |z| \leq \frac{1}{4},$$

and

$$\rho(z) = 0 \quad \text{for } |z| \geq \frac{1}{2}.$$

Set $\psi(z) := \rho(z) - \rho(2z)$. Then

$$(4.34) \quad \text{supp } \psi \subset \left\{ z \in \mathbb{R}^N : \frac{1}{8} \leq |z| \leq \frac{1}{2} \right\}.$$

Moreover, for every $z \neq 0$,

$$(4.35) \quad \sum_{j \in \mathbb{Z}} \psi(2^j z) = 1.$$

Indeed, for $m, n \in \mathbb{N}$,

$$\sum_{j=-m}^n \psi(2^j z) = \rho(2^{-m} z) - \rho(2^{n+1} z),$$

and (4.35) follows by letting $m, n \rightarrow \infty$.

Let $j \in \mathbb{Z}$. For almost every $u \in \Omega$, there is a unique cube $Q \in \mathbb{D}_{\Omega, j}^0$ containing u . If $I := \Phi^{-1}(Q) \in \mathbb{D}_{\Gamma, j}$, then $\ell(I) = 2^{-j}$. Hence, whenever $d(x, y) > 0$, we have $u \neq v$ by (4.33), and therefore

$$(4.36) \quad K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(x, y) = \sum_{j \in \mathbb{Z}} \sum_{I \in \mathbb{D}_{\Gamma, j}} \chi_{\sigma I}(x) K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(x, y) \psi \left(\frac{\Phi(\sigma^{-1} x) - \Phi(\tau^{-1} y)}{\ell(I)} \right).$$

For every fixed pair (x, y) with $d(x, y) > 0$, only finitely many values of j occur in (4.36).

We next record the localization of the y -variable. Let

$$I = \Phi^{-1}(Q), \quad \ell := \ell(I),$$

and suppose that $u \in Q$ and

$$\psi \left(\frac{u - v}{\ell} \right) \neq 0.$$

By (4.34),

$$|u - v| \leq \frac{\ell}{2}.$$

Let c_Q denote the center of Q . Since $u \in Q$, for every $1 \leq j \leq N$,

$$|u_j - (c_Q)_j| \leq \frac{\ell}{2}.$$

Moreover,

$$|u_j - v_j| \leq |u - v| \leq \frac{\ell}{2}.$$

Hence

$$|v_j - (c_Q)_j| \leq |v_j - u_j| + |u_j - (c_Q)_j| \leq \ell$$

for every j . Thus $v \in (2Q) \cap \Omega$, which is equivalent to

$$(4.37) \quad y \in \tau I^*.$$

We show that

$$(4.38) \quad w(I^*) \simeq w(I).$$

Put

$$z_I := \Phi^{-1}(c_Q) \in \Gamma.$$

By (3.8), there exist constants $c_1, C_1 > 0$, independent of I , such that

$$B_\Gamma(z_I, c_1 \ell) \subset I \subset I^* \subset B_\Gamma(z_I, C_1 \ell).$$

Indeed, the first inclusion follows by taking $c_1 > 0$ sufficiently small, whereas the last one follows from

$$(2Q) \cap \Omega \subset B(c_Q, \sqrt{N} \ell)$$

and the boundedness of Φ^{-1} . By the doubling property,

$$w(I^*) \leq w(B_\Gamma(z_I, C_1 \ell)) \lesssim w(B_\Gamma(z_I, c_1 \ell)) \leq w(I).$$

Since $I \subset I^*$, the reverse inequality is immediate, and (4.38) follows.

Step 2.

Fix $\sigma, \tau \in G$ and $I = \Phi^{-1}(Q) \in \mathbb{D}_\Gamma$, and write $\ell := \ell(I)$. Introduce the normalized coordinates

$$a := \frac{u - c_Q}{\ell}, \quad b := \frac{v - c_Q}{\ell}.$$

If $u \in Q$, then $a \in [-1/2, 1/2]^N$. If $v \in (2Q) \cap \Omega$, then b belongs to

$$B_Q := \{b \in [-1, 1]^N : c_Q + \ell b \in \Omega\}.$$

Write

$$Q = \prod_{i=1}^N [m_i \ell, (m_i + 1) \ell),$$

where

$$m_i \in \mathbb{Z}_+ \quad (1 \leq i \leq N_0), \quad m_i \in \mathbb{Z} \quad (N_0 < i \leq N).$$

For $1 \leq i \leq N_0$, the i -th component of B_Q is

$$\begin{cases} [-1/2, 1], & m_i = 0, \\ [-1, 1], & m_i \geq 1, \end{cases}$$

whereas for $i > N_0$ it is always $[-1, 1]$. Hence the rectangles $[-1/2, 1/2]^N \times B_Q$ belong to a finite family.

Consequently, for every Q we may choose $\chi_Q \in C_c^\infty((-2, 2)^N \times (-2, 2)^N)$ such that

$$\chi_Q = 1 \quad \text{on } [-1/2, 1/2]^N \times B_Q,$$

and, for some sufficiently small fixed $\varepsilon_0 > 0$,

$$\text{supp } \chi_Q \subset \{(a, b) : \text{dist}((a, b), [-1/2, 1/2]^N \times B_Q) < \varepsilon_0\}.$$

Since only finitely many such rectangles occur, these functions may be chosen so that, for every integer $M \geq 0$,

$$(4.39) \quad \sup_Q \|\chi_Q\|_{C^M(\mathbb{R}^{2N})} < \infty.$$

We choose ε_0 sufficiently small so that

$$(4.40) \quad d(\sigma\Phi^{-1}(c_Q + \ell a), \tau\Phi^{-1}(c_Q + \ell b)) \simeq \ell$$

whenever

$$(a, b) \in \text{supp } \chi_Q \quad \text{and} \quad a - b \in \text{supp } \psi.$$

Indeed, choose

$$(a_0, b_0) \in [-1/2, 1/2]^N \times B_Q$$

such that

$$|a - a_0| + |b - b_0| \lesssim \varepsilon_0.$$

Since $a - b \in \text{supp } \psi$,

$$|a - b| \geq \frac{1}{8}.$$

Thus, after decreasing ε_0 if necessary,

$$|a_0 - b_0| \geq \frac{1}{16}.$$

Set

$$x'_0 := \Phi^{-1}(c_Q + \ell a_0), \quad y'_0 := \Phi^{-1}(c_Q + \ell b_0).$$

Since

$$c_Q + \ell a_0, \quad c_Q + \ell b_0 \in \Omega,$$

we have $x'_0, y'_0 \in \Gamma$. Hence, by (2.4) and (3.8),

$$d(\sigma x'_0, \tau y'_0) = d(x'_0, y'_0) = \|x'_0 - y'_0\| \gtrsim \ell |a_0 - b_0| \gtrsim \ell.$$

Moreover, directly from the definition of d ,

$$(4.41) \quad |d(x, y) - d(x', y')| \leq \|x - x'\| + \|y - y'\|.$$

Since Φ^{-1} is bounded,

$$\|\Phi^{-1}(c_Q + \ell a) - \Phi^{-1}(c_Q + \ell a_0)\| \lesssim \varepsilon_0 \ell,$$

and similarly for b and b_0 . Therefore, (4.41) gives the lower bound in (4.40) if ε_0 is sufficiently small.

For the reverse estimate, we use the G -invariance of the orbit distance in both variables:

$$d(\sigma x, \tau y) = d(x, y), \quad x, y \in \mathbb{R}^N.$$

Hence

$$\begin{aligned} & d(\sigma \Phi^{-1}(c_Q + \ell a), \tau \Phi^{-1}(c_Q + \ell b)) \\ &= d(\Phi^{-1}(c_Q + \ell a), \Phi^{-1}(c_Q + \ell b)) \\ &\leq \|\Phi^{-1}(c_Q + \ell a) - \Phi^{-1}(c_Q + \ell b)\| \\ &= \|\Phi^{-1}(\ell(a - b))\| \lesssim \ell |a - b| \lesssim \ell, \end{aligned}$$

where in the last step we used $|a - b| \leq 1/2$ on $\text{supp } \psi$. Set

$$X := \sigma \Phi^{-1}(c_Q + \ell a), \quad Y := \tau \Phi^{-1}(c_Q + \ell b).$$

For (a, b) in the above support, X and Y lie within distance $C\ell$ of σz_I and τz_I , respectively. Combining (4.40), the doubling estimate, the G -invariance of dw , and (3.15), we obtain

$$w(B(X, d(X, Y))) \simeq w(B(z_I, \ell)) \simeq w(I)$$

and similarly

$$w(B(Y, d(X, Y))) \simeq w(I).$$

Consequently,

$$(4.42) \quad V(X, Y, d(X, Y)) \simeq w(I).$$

By Lemma 4.2, for every pair of multi-indices α, β ,

$$\left| \partial_x^\alpha \partial_y^\beta K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(X, Y) \right| \lesssim \frac{d(X, Y)^{1-|\alpha|-|\beta|}}{V(X, Y, d(X, Y))}.$$

Combining this with (4.40) and (4.42), we obtain

$$(4.43) \quad \left| \partial_x^\alpha \partial_y^\beta K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(X, Y) \right| \lesssim_{\alpha, \beta} \frac{\ell^{1-|\alpha|-|\beta|}}{w(I)}.$$

Step 3.

Define

$$(4.44) \quad H_{\sigma, \tau, I}(a, b) := \frac{w(I)}{\ell(I)} \chi_Q(a, b) \psi(a - b) K_{(-\Delta_{\text{Dunkl}})^{-1/2}} \left(\sigma \Phi^{-1}(c_Q + \ell(I)a), \tau \Phi^{-1}(c_Q + \ell(I)b) \right).$$

We claim that, for every integer $M \geq 0$,

$$(4.45) \quad \sup_{\sigma, \tau \in G} \sup_{I \in \mathbb{D}_\Gamma} \|H_{\sigma, \tau, I}\|_{C^M(\mathbb{R}^{2N})} < \infty.$$

Indeed, if altogether k derivatives fall on the kernel term, the chain rule produces a factor bounded by a constant multiple of $\ell(I)^k$. By (4.43), the corresponding contribution is bounded by

$$\frac{w(I)}{\ell(I)} \ell(I)^k \frac{\ell(I)^{1-k}}{w(I)} \lesssim 1.$$

The derivatives falling on χ_Q and ψ are uniformly bounded by (4.39) and the fact that ψ is fixed. The product rule proves (4.45).

Since $H_{\sigma, \tau, I}$ is compactly supported in the interior of $[-2, 2]^N \times [-2, 2]^N$, its 4-periodic extension is smooth. Hence

$$(4.46) \quad H_{\sigma, \tau, I}(a, b) = \sum_{p, q \in \mathbb{Z}^N} b_{p, q, I}^{\sigma, \tau} e^{\frac{\pi i}{2} p \cdot a} e^{\frac{\pi i}{2} q \cdot b}.$$

Repeated integration by parts and (4.45) yield, for every $M > 0$,

$$(4.47) \quad \sup_{\sigma, \tau \in G} \sup_{I \in \mathbb{D}_\Gamma} |b_{p, q, I}^{\sigma, \tau}| \leq C_M (1 + |p| + |q|)^{-M}.$$

When $x \in \sigma I$ and $y \in \tau I^*$, the corresponding normalized variables satisfy

$$a \in [-1/2, 1/2]^N, \quad b \in B_Q,$$

and therefore $\chi_Q(a, b) = 1$. Thus

$$(4.48) \quad \begin{aligned} & K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(x, y) \psi \left(\frac{\Phi(\sigma^{-1}x) - \Phi(\tau^{-1}y)}{\ell(I)} \right) \\ &= \frac{\ell(I)}{w(I)} \sum_{p, q \in \mathbb{Z}^N} b_{p, q, I}^{\sigma, \tau} \exp \left(\frac{\pi i}{2\ell(I)} p \cdot (\Phi(\sigma^{-1}x) - c_Q) \right) \exp \left(\frac{\pi i}{2\ell(I)} q \cdot (\Phi(\tau^{-1}y) - c_Q) \right). \end{aligned}$$

Define

$$(4.49) \quad c_{p, q} := \sup_{\substack{\sigma, \tau \in G \\ I \in \mathbb{D}_\Gamma}} \left\{ |b_{p, q, I}^{\sigma, \tau}| \left(\frac{w(I^*)}{w(I)} \right)^{1/2} \right\}.$$

By (4.38) and (4.47), for every $M > 0$,

$$c_{p, q} \leq C_M (1 + |p| + |q|)^{-M}.$$

Thus (4.29) holds.

Define

$$a_{p,q,I}^{\sigma,\tau} := \begin{cases} \frac{b_{p,q,I}^{\sigma,\tau}}{c_{p,q}} \left(\frac{w(I^*)}{w(I)} \right)^{1/2}, & c_{p,q} > 0, \\ 0, & c_{p,q} = 0. \end{cases}$$

Then

$$(4.50) \quad |a_{p,q,I}^{\sigma,\tau}| \leq 1,$$

and

$$(4.51) \quad b_{p,q,I}^{\sigma,\tau} = c_{p,q} a_{p,q,I}^{\sigma,\tau} \left(\frac{w(I)}{w(I^*)} \right)^{1/2}.$$

Step 4.

If $\|f\|_{L^r(\sigma I, dw)} > 0$, define

$$(4.52) \quad e_{p,q,I}^{\sigma,\tau}(x) := a_{p,q,I}^{\sigma,\tau} w(I)^{\frac{1}{r}-\frac{1}{2}} \frac{f(x) \chi_{\sigma I}(x)}{\|f\|_{L^r(\sigma I, dw)}} \exp\left(\frac{\pi i}{2\ell(I)} p \cdot (\Phi(\sigma^{-1}x) - c_Q)\right).$$

If $\|f\|_{L^r(\sigma I, dw)} = 0$, set $e_{p,q,I}^{\sigma,\tau} := 0$. Define

$$(4.53) \quad h_{p,q,I}^{\sigma,\tau}(y) := w(I^*)^{-1/2} \chi_{\tau I^*}(y) \exp\left(\frac{\pi i}{2\ell(I)} q \cdot (\Phi(\tau^{-1}y) - c_Q)\right).$$

By construction,

$$\text{supp } e_{p,q,I}^{\sigma,\tau} \subset \sigma I, \quad \text{supp } h_{p,q,I}^{\sigma,\tau} \subset \tau I^*.$$

Since dw is G -invariant,

$$w(\sigma I) = w(I), \quad w(\tau I^*) = w(I^*).$$

By (4.50), if $\|f\|_{L^r(\sigma I, dw)} > 0$, then

$$\|e_{p,q,I}^{\sigma,\tau}\|_{L^r(dw)} \leq w(I)^{\frac{1}{r}-\frac{1}{2}} \frac{\|f \chi_{\sigma I}\|_{L^r(dw)}}{\|f\|_{L^r(\sigma I, dw)}} = w(I)^{\frac{1}{r}-\frac{1}{2}}.$$

The same estimate is trivial when $\|f\|_{L^r(\sigma I, dw)} = 0$. Hence

$$\|e_{p,q,I}^{\sigma,\tau}\|_{L^r(\mathbb{R}^N, dw)} \leq w(I)^{\frac{1}{r}-\frac{1}{2}}.$$

Similarly,

$$\|h_{p,q,I}^{\sigma,\tau}\|_{L^r(dw)} = w(I^*)^{-1/2} w(I^*)^{1/r} = w(I^*)^{\frac{1}{r}-\frac{1}{2}}.$$

Since $I \subset I^*$, this yields

$$w(I^*)^{\frac{1}{r}-\frac{1}{2}} \leq w(I)^{\frac{1}{r}-\frac{1}{2}}.$$

Therefore,

$$(4.54) \quad \|e_{p,q,I}^{\sigma,\tau}\|_{L^r(\mathbb{R}^N, dw)} \leq w(I)^{\frac{1}{r}-\frac{1}{2}}, \quad \|h_{p,q,I}^{\sigma,\tau}\|_{L^r(\mathbb{R}^N, dw)} \leq w(I)^{\frac{1}{r}-\frac{1}{2}},$$

which is precisely (4.31).

Step 5.

Suppose first that $\|f\|_{L^r(\sigma I, dw)} > 0$. By the definition of $\lambda_{\sigma I}(f)$, (4.52), (4.53), and (4.51), for $x \in \sigma I$ and $y \in \tau I^*$ we have

$$c_{p,q} \lambda_{\sigma I}(f) e_{p,q,I}^{\sigma,\tau}(x) h_{p,q,I}^{\sigma,\tau}(y)$$

$$\begin{aligned}
&= c_{p,q} \left[\ell(I) w(I)^{-1/r} \|f\|_{L^r(\sigma I, dw)} \right] \\
&\quad \times a_{p,q,I}^{\sigma,\tau} w(I)^{\frac{1}{r}-\frac{1}{2}} \frac{f(x)}{\|f\|_{L^r(\sigma I, dw)}} w(I^*)^{-1/2} \\
&\quad \times \exp \left(\frac{\pi i}{2\ell(I)} p \cdot (\Phi(\sigma^{-1}x) - c_Q) \right) \\
&\quad \times \exp \left(\frac{\pi i}{2\ell(I)} q \cdot (\Phi(\tau^{-1}y) - c_Q) \right) \\
&= c_{p,q} a_{p,q,I}^{\sigma,\tau} \ell(I) w(I)^{-1/2} w(I^*)^{-1/2} f(x) \\
&\quad \times \exp \left(\frac{\pi i}{2\ell(I)} p \cdot (\Phi(\sigma^{-1}x) - c_Q) \right) \\
&\quad \times \exp \left(\frac{\pi i}{2\ell(I)} q \cdot (\Phi(\tau^{-1}y) - c_Q) \right).
\end{aligned}$$

By (4.51),

$$c_{p,q} a_{p,q,I}^{\sigma,\tau} w(I)^{-1/2} w(I^*)^{-1/2} = b_{p,q,I}^{\sigma,\tau} w(I)^{-1}.$$

Therefore,

$$\begin{aligned}
&c_{p,q} \lambda_{\sigma,I}(f) e_{p,q,I}^{\sigma,\tau}(x) h_{p,q,I}^{\sigma,\tau}(y) \\
&= b_{p,q,I}^{\sigma,\tau} \frac{\ell(I)}{w(I)} f(x) \exp \left(\frac{\pi i}{2\ell(I)} p \cdot (\Phi(\sigma^{-1}x) - c_Q) \right) \exp \left(\frac{\pi i}{2\ell(I)} q \cdot (\Phi(\tau^{-1}y) - c_Q) \right).
\end{aligned}$$

Summing over p, q and using (4.48), we obtain

$$\begin{aligned}
&f(x) K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(x, y) \psi \left(\frac{\Phi(\sigma^{-1}x) - \Phi(\tau^{-1}y)}{\ell(I)} \right) \\
(4.55) \quad &= \sum_{p,q \in \mathbb{Z}^N} c_{p,q} \lambda_{\sigma,I}(f) e_{p,q,I}^{\sigma,\tau}(x) h_{p,q,I}^{\sigma,\tau}(y).
\end{aligned}$$

If $\|f\|_{L^r(\sigma I, dw)} = 0$, then $f = 0$ almost everywhere on σI , while $e_{p,q,I}^{\sigma,\tau} = 0$. Hence (4.55) remains valid for almost every $x \in \sigma I$.

If $x \in \sigma I$ but $y \notin \tau I^*$, then (4.37) implies that the left-hand side of (4.55) vanishes, while $h_{p,q,I}^{\sigma,\tau}(y) = 0$. Thus (4.55) holds for almost every $(x, y) \in \sigma I \times \tau(\Gamma)$.

We next justify the absolute convergence needed to rearrange the series. Fix σ, τ and almost every (x, y) such that

$$x \in \sigma(\Gamma)^\circ, \quad y \in \tau(\Gamma)^\circ, \quad d(x, y) > 0.$$

If a term indexed by I is nonzero, then

$$x \in \sigma I, \quad y \in \tau I^*.$$

The geometry of I and I^* , together with (2.4) and (3.8), gives

$$d(x, y) \lesssim \ell(I).$$

For each fixed generation there is at most one $I \in \mathbb{D}_\Gamma$ containing $\sigma^{-1}x$.

Moreover, from

$$B_\Gamma(z_I, c_1 \ell(I)) \subset I$$

and the volume estimates in the text,

$$w(I) \gtrsim \ell(I)^{\mathbf{N}}.$$

Using the definitions above and (4.47), we obtain, for every $M > 0$,

$$\begin{aligned} c_{p,q} \lambda_{\sigma,I}(f) \left| e_{p,q,I}^{\sigma,\tau}(x) h_{p,q,I}^{\sigma,\tau}(y) \right| \\ \lesssim |f(x)| (1 + |p| + |q|)^{-M} \frac{\ell(I)}{w(I)} \\ \lesssim |f(x)| (1 + |p| + |q|)^{-M} \ell(I)^{1-\mathbf{N}}. \end{aligned}$$

Since $\mathbf{N} > 1$, the sum over all dyadic scales satisfying

$$\ell(I) \gtrsim d(x, y)$$

converges. Choosing $M > 2N$ also gives

$$\sum_{p,q \in \mathbb{Z}^N} (1 + |p| + |q|)^{-M} < \infty.$$

Hence the relevant series is absolutely convergent for almost every such (x, y) .

Summing (4.55) first over $I \in \mathbb{D}_{\Gamma,j}$ and then over $j \in \mathbb{Z}$, and using (4.36), gives

$$f(x) K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(x, y) = \sum_{p,q \in \mathbb{Z}^N} c_{p,q} \sum_{I \in \mathbb{D}_{\Gamma}} \lambda_{\sigma,I}(f) e_{p,q,I}^{\sigma,\tau}(x) h_{p,q,I}^{\sigma,\tau}(y)$$

for almost every

$$x \in \sigma(\Gamma)^{\circ}, \quad y \in \tau(\Gamma)^{\circ}, \quad d(x, y) > 0.$$

Finally, by the Weyl chamber decomposition established above in the text, the interiors of the chambers $\sigma(\Gamma)$ cover $\mathbb{R}^N$ up to the reflecting hyperplanes, which form a dw -null set. Summing over $\sigma, \tau \in G$, we obtain

$$f(x) K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(x, y) = \sum_{\sigma, \tau \in G} \sum_{p,q \in \mathbb{Z}^N} c_{p,q} \sum_{I \in \mathbb{D}_{\Gamma}} \lambda_{\sigma,I}(f) e_{p,q,I}^{\sigma,\tau}(x) h_{p,q,I}^{\sigma,\tau}(y)$$

for almost every (x, y) with $d(x, y) > 0$.

Thus (4.32) holds, and the proof is complete. $\square$

Lemma 4.4. *Let $1 < s < \infty$, and let $\mathbb{D}^0$ be the standard dyadic grid in $\mathbb{R}^N$. Suppose that $F \in L^s(\mathbb{R}^N)$. For every $Q \in \mathbb{D}^0$, set*

$$b_Q := |Q|^{\frac{1}{s}-1} \int_Q |F(x)| dx = |Q|^{-1/s'} \int_Q |F(x)| dx,$$

Then

$$(4.56) \quad \#\{Q \in \mathbb{D}^0 : b_Q > t\} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s, \quad t > 0.$$

Equivalently,

$$\left\| \{b_Q\}_{Q \in \mathbb{D}^0} \right\|_{\ell^{s,\infty}} \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)}.$$

Proof. Let $\mathcal{E} \subset \mathbb{D}^0$ be an arbitrary finite collection of dyadic cubes, and define

$$G_{\mathcal{E}}(x) := \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x).$$

We first claim that

$$(4.57) \quad \|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)} \lesssim_{N,s} (\#\mathcal{E})^{1/s'}.$$

Indeed, fix

$$x \in \bigcup_{Q \in \mathcal{E}} Q.$$

Set

$$\mathcal{E}_x := \{Q \in \mathcal{E} : x \in Q\}.$$

Since $\mathcal{E}$ is finite, there exists a cube $Q_x \in \mathcal{E}_x$ having minimal volume. Since any two cubes in the same dyadic grid which contain the same point are nested, every $Q \in \mathcal{E}_x$ contains Q_x . Thus the cubes in $\mathcal{E}_x$ belong to the chain of dyadic ancestors of Q_x .

Denote by $Q_x^{(k)}$ the k -th dyadic ancestor of Q_x , so that

$$|Q_x^{(k)}| = 2^{kN} |Q_x|, \quad k \geq 0.$$

It follows that

$$\begin{aligned} G_{\mathcal{E}}(x) &= \sum_{Q \in \mathcal{E}_x} |Q|^{-1/s'} \\ &\leq \sum_{k=0}^{\infty} |Q_x^{(k)}|^{-1/s'} \\ &= |Q_x|^{-1/s'} \sum_{k=0}^{\infty} 2^{-kN/s'} \\ &\lesssim_{N,s} |Q_x|^{-1/s'}. \end{aligned}$$

For each $Q \in \mathcal{E}$, define

$$E_Q := \left\{ x \in \bigcup_{P \in \mathcal{E}} P : Q_x = Q \right\}.$$

The sets $\{E_Q\}_{Q \in \mathcal{E}}$ are pairwise disjoint and satisfy

$$E_Q \subset Q.$$

Therefore,

$$\begin{aligned} \|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)}^{s'} &= \int_{\mathbb{R}^N} G_{\mathcal{E}}(x)^{s'} dx \\ &\lesssim_{N,s} \sum_{Q \in \mathcal{E}} \int_{E_Q} |Q|^{-1} dx \\ &= \sum_{Q \in \mathcal{E}} \frac{|E_Q|}{|Q|} \end{aligned}$$

$$\leq \sum_{Q \in \mathcal{E}} 1 = \#\mathcal{E}.$$

Taking the s' -th root proves (4.57).

We now estimate the sum of $\{b_Q\}_{Q \in \mathcal{E}}$. By Hölder's inequality and (4.57),

$$\begin{aligned} \sum_{Q \in \mathcal{E}} b_Q &= \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \int_Q |F(x)| dx \\ &= \int_{\mathbb{R}^N} |F(x)| \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x) dx \\ &= \int_{\mathbb{R}^N} |F(x)| G_{\mathcal{E}}(x) dx \\ &\leq \|F\|_{L^s(\mathbb{R}^N)} \|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)} \\ &\lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'}. \end{aligned}$$

Now fix $t > 0$ and let

$$\mathcal{E} \subset \{Q \in \mathbb{D}^0 : b_Q > t\}$$

be an arbitrary finite subcollection. Since $b_Q > t$ for every $Q \in \mathcal{E}$, we have

$$t \#\mathcal{E} < \sum_{Q \in \mathcal{E}} b_Q \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'}.$$

If $\mathcal{E} \neq \emptyset$, dividing by $(\#\mathcal{E})^{1/s'}$ and using

$$1 - \frac{1}{s'} = \frac{1}{s},$$

we obtain

$$(\#\mathcal{E})^{1/s} \lesssim_{N,s} t^{-1} \|F\|_{L^s(\mathbb{R}^N)}.$$

Consequently,

$$\#\mathcal{E} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s.$$

Since this estimate holds for every finite subcollection

$$\mathcal{E} \subset \{Q \in \mathbb{D}^0 : b_Q > t\},$$

it follows that

$$\#\{Q \in \mathbb{D}^0 : b_Q > t\} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s.$$

This proves (4.56), and hence also the stated weak ℓ^s estimate. $\square$

Lemma 4.5. *Assume that $N > 2$ and $2 < r < N$. Then, for every $\sigma \in G$ and every $f \in L^N(\mathbb{R}^N, dx)$, we have $f \in L_{\text{loc}}^r(\mathbb{R}^N, dw)$ and*

$$(4.58) \quad \left\| \{\lambda_{\sigma, I}(f)\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{N, \infty}} \lesssim \|f\|_{L^N(\mathbb{R}^N, dx)}.$$

The implicit constant is independent of σ and f .

Proof. We first note that $f \in L^r_{\text{loc}}(\mathbb{R}^N, dw)$. Indeed, the density of dw with respect to Lebesgue measure is $\prod_{v \in R_+} |\langle x, v \rangle|^{2k(v)}$, which is bounded on every compact subset of $\mathbb{R}^N$. Hence, if $E \subset \mathbb{R}^N$ is bounded, then, since $r < N$,

$$\int_E |f(x)|^r dw(x) \lesssim_E \int_E |f(x)|^r dx \leq |E|^{1-\frac{r}{N}} \|f\|_{L^N(\mathbb{R}^N, dx)}^r < \infty.$$

Thus $\lambda_{\sigma, I}(f)$ in (4.19) is well defined.

Fix $\sigma \in G$. Recall that $\Omega = \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}$ and, by Lemma 3.9,

$$(4.59) \quad d\mu(u, z) = \phi(u) du dz, \quad \phi(u) = |\det(\Phi)|^{-1} \prod_{v \in R_+} \left(\sum_{i=1}^{N_0} a_{v,i} u_i \right)^{2k(v)},$$

where $a_{v,i} \geq 0$. In particular, ϕ is nondecreasing in each of the variables $u_1, \dots, u_{N_0}$ and is independent of the remaining $N - N_0$ variables.

Let $I = \Phi^{-1}(Q) \in \mathbb{D}_\Gamma$, where

$$Q = \prod_{i=1}^N [m_i \ell, (m_i + 1) \ell] \subset \Omega, \quad \ell = \ell(Q) = \ell(I),$$

with

$$m_i \in \mathbb{Z}_+, \quad 1 \leq i \leq N_0,$$

and

$$m_i \in \mathbb{Z}, \quad N_0 < i \leq N.$$

Define $u^+ := ((m_1 + 1)\ell, \dots, (m_{N_0} + 1)\ell)$ and

$$Q^+ := \prod_{i=1}^{N_0} \left[\left(m_i + \frac{1}{2} \right) \ell, (m_i + 1) \ell \right) \times \prod_{i=N_0+1}^N [m_i \ell, (m_i + 1) \ell).$$

Then

$$|Q^+| = 2^{-N_0} |Q|.$$

For every $(u, z) \in Q^+$ and every $1 \leq i \leq N_0$,

$$u_i \geq \left(m_i + \frac{1}{2} \right) \ell \geq \frac{1}{2} (m_i + 1) \ell.$$

Since $a_{v,i} \geq 0$, it follows that, for every $v \in R_+$,

$$\sum_{i=1}^{N_0} a_{v,i} u_i \geq \frac{1}{2} \sum_{i=1}^{N_0} a_{v,i} (m_i + 1) \ell.$$

Consequently,

$$\phi(u) \geq 2^{-2 \sum_{v \in R_+} k(v)} \phi(u^+).$$

Since

$$2 \sum_{v \in R_+} k(v) = \sum_{v \in R} k(v) = \mathbf{N} - N,$$

we obtain

$$\phi(u) \geq 2^{-(\mathbf{N}-N)} \phi(u^+), \quad (u, z) \in Q^+.$$

Therefore,

$$\begin{aligned}
\mu(Q) &= \int_Q \phi(u) du dz \\
&\geq \int_{Q^+} \phi(u) du dz \\
&\geq 2^{-(N-N)} |Q^+| \phi(u^+) \\
&= 2^{-(N-N+N_0)} |Q| \phi(u^+).
\end{aligned}$$

On the other hand, by the coordinatewise monotonicity of ϕ ,

$$\operatorname{ess\,sup}_{(u,z) \in Q} \phi(u) \leq \phi(u^+).$$

It follows that

$$(4.60) \quad \operatorname{ess\,sup}_Q \phi \lesssim \frac{\mu(Q)}{|Q|}.$$

Since dw is G -invariant and

$$\mu(Q) = w(\Phi^{-1}(Q)) = w(I),$$

we have

$$\begin{aligned}
\frac{1}{w(I)} \int_{\sigma I} |f(x)|^r dw(x) &= \frac{1}{w(I)} \int_I |f(\sigma x)|^r dw(x) \\
&= \frac{1}{\mu(Q)} \int_Q |f(\sigma \Phi^{-1}(\xi))|^r d\mu(\xi) \\
&= \frac{1}{\mu(Q)} \int_Q |f(\sigma \Phi^{-1}(\xi))|^r \phi(\xi) d\xi \\
&\leq \frac{\operatorname{ess\,sup}_Q \phi}{\mu(Q)} \int_Q |f(\sigma \Phi^{-1}(\xi))|^r d\xi \\
&\lesssim \frac{1}{|Q|} \int_Q |f(\sigma \Phi^{-1}(\xi))|^r d\xi,
\end{aligned}$$

where we used (4.60) in the last inequality. Hence, by the definition (4.19),

$$\begin{aligned}
\lambda_{\sigma, I}(f)^r &= \ell(I)^r \frac{1}{w(I)} \int_{\sigma I} |f(x)|^r dw(x) \\
(4.61) \quad &\lesssim \frac{\ell(Q)^r}{|Q|} \int_Q |f(\sigma \Phi^{-1}(\xi))|^r d\xi.
\end{aligned}$$

Set

$$s := \frac{N}{r}.$$

Since $r < N$, we have $s > 1$. Moreover,

$$|Q| = \ell(Q)^N,$$

and hence

$$\frac{\ell(Q)^r}{|Q|} = |Q|^{\frac{r}{N}-1} = |Q|^{\frac{1}{s}-1}.$$

Define

$$F_\sigma(\xi) := \begin{cases} |f(\sigma\Phi^{-1}(\xi))|^r, & \xi \in \Omega, \\ 0, & \xi \notin \Omega. \end{cases}$$

Then (4.61) gives

$$(4.62) \quad \lambda_{\sigma,I}(f)^r \lesssim |Q|^{\frac{1}{s}-1} \int_Q F_\sigma(\xi) d\xi.$$

We next verify that $F_\sigma \in L^s(\mathbb{R}^N)$. Since $rs = N$, a change of variables $\xi = \Phi(x)$ gives

$$\begin{aligned} \|F_\sigma\|_{L^s(\mathbb{R}^N)}^s &= \int_\Omega |f(\sigma\Phi^{-1}(\xi))|^{rs} d\xi \\ &= \int_\Omega |f(\sigma\Phi^{-1}(\xi))|^N d\xi \\ &= |\det(\Phi)| \int_\Gamma |f(\sigma x)|^N dx \\ &= |\det(\Phi)| \int_{\sigma\Gamma} |f(x)|^N dx \\ &\leq |\det(\Phi)| \|f\|_{L^N(\mathbb{R}^N, dx)}^N < \infty, \end{aligned}$$

where in the fourth line we used the fact that σ is orthogonal. Thus $F_\sigma \in L^s(\mathbb{R}^N)$.

Since $\mathbb{D}_\Omega^0$ is a subfamily of the standard dyadic system $\mathbb{D}^0$, we may apply Lemma 4.4 to F_σ . Together with (4.62), this gives, for every $a > 0$,

$$\begin{aligned} \#\{I \in \mathbb{D}_\Gamma : \lambda_{\sigma,I}(f) > a\} &\leq \#\left\{Q \in \mathbb{D}_\Omega^0 : |Q|^{\frac{1}{s}-1} \int_Q F_\sigma(\xi) d\xi \gtrsim a^r\right\} \\ &\leq \#\left\{Q \in \mathbb{D}^0 : |Q|^{\frac{1}{s}-1} \int_Q F_\sigma(\xi) d\xi \gtrsim a^r\right\} \\ &\lesssim (a^r)^{-s} \|F_\sigma\|_{L^s(\mathbb{R}^N)}^s \\ &= a^{-rs} \|F_\sigma\|_{L^s(\mathbb{R}^N)}^s \\ &\lesssim a^{-N} \|f\|_{L^N(\mathbb{R}^N, dx)}^N, \end{aligned}$$

where in the last line we used $rs = N$ and the estimate for $\|F_\sigma\|_{L^s(\mathbb{R}^N)}^s$ obtained above. Therefore,

$$\#\{I \in \mathbb{D}_\Gamma : \lambda_{\sigma,I}(f) > a\} \lesssim \left(\frac{\|f\|_{L^N(\mathbb{R}^N, dx)}}{a} \right)^N, \quad a > 0.$$

By the definition of the weak ℓ^N quasi-norm, this is exactly

$$\left\| \{\lambda_{\sigma,I}(f)\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{N,\infty}} \lesssim \|f\|_{L^N(\mathbb{R}^N, dx)}.$$

All the implicit constants above depend only on the fixed structural parameters and are independent of $\sigma \in G$ and f . This completes the proof. $\square$

Lemma 4.6. Assume that $N > 2$ and $2 < r < N$. Let

$$\left\{ e_{p,q,I}^{\sigma,\tau} \right\}_{p,q \in \mathbb{Z}^N, I \in \mathbb{D}_\Gamma, \sigma, \tau \in G} \quad \text{and} \quad \left\{ h_{p,q,I}^{\sigma,\tau} \right\}_{p,q \in \mathbb{Z}^N, I \in \mathbb{D}_\Gamma, \sigma, \tau \in G}$$

be the families obtained from Lemma 4.3 with this choice of r . For every

$$f \in L^N(\mathbb{R}^N, dx), \quad \sigma, \tau \in G, \quad p, q \in \mathbb{Z}^N,$$

define

$$A_{\sigma, \tau, p, q}(f) := \sum_{I \in \mathbb{D}_\Gamma} \lambda_{\sigma, I}(f) \left\langle \cdot, \overline{h_{p, q, I}^{\sigma, \tau}} \right\rangle_{L^2(\mathbb{R}^N, dw)} e_{p, q, I}^{\sigma, \tau}.$$

Then

$$A_{\sigma, \tau, p, q}(f) \in S^{N, \infty}(L^2(\mathbb{R}^N, dw))$$

and

$$(4.63) \quad \|A_{\sigma, \tau, p, q}(f)\|_{S^{N, \infty}(L^2(\mathbb{R}^N, dw))} \lesssim \|f\|_{L^N(\mathbb{R}^N, dx)},$$

where the implicit constant is independent of σ, τ, p, q , and f .

Proof. By Lemma 4.5,

$$(4.64) \quad \|\{\lambda_{\sigma, I}(f)\}_{I \in \mathbb{D}_\Gamma}\|_{\ell^{N, \infty}} \lesssim \|f\|_{L^N(\mathbb{R}^N, dx)},$$

uniformly in $\sigma \in G$.

Fix

$$\sigma, \tau \in G, \quad p, q \in \mathbb{Z}^N.$$

For $\gamma \in G$, define

$$U_\gamma : L^2(\Gamma, dw) \longrightarrow L^2(\gamma\Gamma, dw)$$

by

$$U_\gamma u(x) := u(\gamma^{-1}x), \quad x \in \gamma\Gamma.$$

Since dw is G -invariant, U_γ is unitary. In fact, the same argument shows that U_γ is an isometry on $L^s(dw)$ for every $1 \leq s \leq \infty$.

For $I \in \mathbb{D}_\Gamma$, define

$$\widetilde{e}_I := U_\sigma^{-1} e_{p, q, I}^{\sigma, \tau}$$

and

$$\widetilde{h}_I := U_\tau^{-1} \overline{h_{p, q, I}^{\sigma, \tau}}.$$

By (4.30),

$$\text{supp } \widetilde{e}_I \subset I, \quad \text{supp } \widetilde{h}_I \subset I^*.$$

We claim that

$$\{\widetilde{e}_I\}_{I \in \mathbb{D}_\Gamma} \quad \text{and} \quad \{\widetilde{h}_I\}_{I \in \mathbb{D}_\Gamma}$$

are NWO sequences.

Indeed, write

$$I = \Phi^{-1}(Q), \quad I^* = \Phi^{-1}((2Q) \cap \Omega).$$

Since Φ and Φ^{-1} are bi-Lipschitz, the set I^* is contained in a fixed geometric enlargement of I , with the enlargement constant independent of I . Thus the support condition in the definition of an NWO sequence is satisfied for both families.

Moreover, since U_σ and U_τ preserve the $L^r(dw)$ norm, (4.31) gives

$$(4.65) \quad \|\tilde{e}_I\|_{L^r(\Gamma, dw)} = \|e_{p,q,I}^{\sigma,\tau}\|_{L^r(\mathbb{R}^N, dw)} \leq w(I)^{\frac{1}{r}-\frac{1}{2}}$$

and

$$(4.66) \quad \|\tilde{h}_I\|_{L^r(\Gamma, dw)} = \|h_{p,q,I}^{\sigma,\tau}\|_{L^r(\mathbb{R}^N, dw)} \leq w(I)^{\frac{1}{r}-\frac{1}{2}}.$$

Since $r > 2$, these are precisely the required L^r normalization conditions for NWO sequences. Therefore

$$\{\tilde{e}_I\}_{I \in \mathbb{D}_\Gamma} \quad \text{and} \quad \{\tilde{h}_I\}_{I \in \mathbb{D}_\Gamma}$$

are NWO sequences.

Consider now the operator on $L^2(\Gamma, dw)$ given by

$$(4.67) \quad \tilde{A}_{\sigma,\tau,p,q}(f) := \sum_{I \in \mathbb{D}_\Gamma} \lambda_{\sigma,I}(f) \langle \cdot, \tilde{h}_I \rangle_{L^2(\Gamma, dw)} \tilde{e}_I.$$

Applying Lemma 2.11 with exponent N , and then using (4.64), we obtain

$$(4.68) \quad \|\tilde{A}_{\sigma,\tau,p,q}(f)\|_{S^{N,\infty}(L^2(\Gamma, dw))} \lesssim \|\{\lambda_{\sigma,I}(f)\}_{I \in \mathbb{D}_\Gamma}\|_{\ell^{N,\infty}} \lesssim \|f\|_{L^N(\mathbb{R}^N, dx)}.$$

It remains to transfer this estimate back to $L^2(\mathbb{R}^N, dw)$.

By (4.30), the range of $A_{\sigma,\tau,p,q}(f)$ is contained in $L^2(\sigma\Gamma, dw)$, whereas the operator vanishes on $L^2(\mathbb{R}^N \setminus \tau\Gamma, dw)$. Thus its nonzero part is an operator

$$L^2(\tau\Gamma, dw) \longrightarrow L^2(\sigma\Gamma, dw).$$

For $g \in L^2(\tau\Gamma, dw)$, put

$$u := U_\tau^{-1}g.$$

Then, using the definition of $\tilde{h}_I$ and the unitarity of U_τ ,

$$\langle g, \overline{h_{p,q,I}^{\sigma,\tau}} \rangle_{L^2(\tau\Gamma, dw)} = \langle U_\tau^{-1}g, U_\tau^{-1}\overline{h_{p,q,I}^{\sigma,\tau}} \rangle_{L^2(\Gamma, dw)} = \langle u, \tilde{h}_I \rangle_{L^2(\Gamma, dw)}.$$

Also,

$$e_{p,q,I}^{\sigma,\tau} = U_\sigma \tilde{e}_I.$$

Consequently,

$$\begin{aligned} A_{\sigma,\tau,p,q}(f)g &= \sum_{I \in \mathbb{D}_\Gamma} \lambda_{\sigma,I}(f) \langle g, \overline{h_{p,q,I}^{\sigma,\tau}} \rangle e_{p,q,I}^{\sigma,\tau} \\ &= U_\sigma \sum_{I \in \mathbb{D}_\Gamma} \lambda_{\sigma,I}(f) \langle u, \tilde{h}_I \rangle \tilde{e}_I \\ &= U_\sigma \tilde{A}_{\sigma,\tau,p,q}(f) U_\tau^{-1}g. \end{aligned}$$

Hence

$$(4.69) \quad A_{\sigma,\tau,p,q}(f) = U_\sigma \tilde{A}_{\sigma,\tau,p,q}(f) U_\tau^{-1}$$

on $L^2(\tau\Gamma, dw)$, with the left-hand side extended by zero on the orthogonal complement of $L^2(\tau\Gamma, dw)$.

Therefore the nonzero singular values of $A_{\sigma,\tau,p,q}(f)$ and $\tilde{A}_{\sigma,\tau,p,q}(f)$ coincide. In particular,

$$\|A_{\sigma,\tau,p,q}(f)\|_{S^{N,\infty}(L^2(\mathbb{R}^N, dw))} = \|\tilde{A}_{\sigma,\tau,p,q}(f)\|_{S^{N,\infty}(L^2(\Gamma, dw))}.$$

Combining this identity with (4.68), we conclude that

$$\|A_{\sigma,\tau,p,q}(f)\|_{S^{N,\infty}(L^2(\mathbb{R}^N,dw))} \lesssim \|f\|_{L^N(\mathbb{R}^N,dx)}.$$

The implicit constant is independent of σ, τ, p, q , and f , which proves (4.63). $\square$

Lemma 4.7. *Assume that $N > 2$. Then there exists a constant $C_{N,R,k} > 0$ such that, for every $f \in L^N(\mathbb{R}^N, dx)$,*

$$M_f(-\Delta_{\text{Dunkl}})^{-1/2} \in S^{N,\infty}(L^2(\mathbb{R}^N, dw))$$

and

$$(4.70) \quad \|M_f(-\Delta_{\text{Dunkl}})^{-1/2}\|_{S^{N,\infty}(L^2(\mathbb{R}^N,dw))} \leq C_{N,R,k} \|f\|_{L^N(\mathbb{R}^N,dx)}.$$

Proof. Choose $2 < r < N$. By Lemma 4.5, $f \in L^r_{\text{loc}}(\mathbb{R}^N, dw)$, and hence Lemma 4.3 is applicable with this choice of r .

Let

$$\{c_{p,q}\}_{p,q \in \mathbb{Z}^N}, \quad \{e_{p,q,I}^{\sigma,\tau}\}, \quad \{h_{p,q,I}^{\sigma,\tau}\}$$

be the coefficients and functions given by Lemma 4.3. For $\sigma, \tau \in G$ and $p, q \in \mathbb{Z}^N$, let

$$A_{\sigma,\tau,p,q}(f) := \sum_{I \in \mathbb{D}_\Gamma} \lambda_{\sigma,I}(f) \left\langle \cdot, \overline{h_{p,q,I}^{\sigma,\tau}} \right\rangle_{L^2(\mathbb{R}^N,dw)} e_{p,q,I}^{\sigma,\tau}.$$

By Lemma 4.6,

$$(4.71) \quad \|A_{\sigma,\tau,p,q}(f)\|_{S^{N,\infty}(L^2(\mathbb{R}^N,dw))} \lesssim \|f\|_{L^N(\mathbb{R}^N,dx)}$$

uniformly in

$$\sigma, \tau \in G, \quad p, q \in \mathbb{Z}^N.$$

On the other hand, by (4.29), for every $M > 0$,

$$c_{p,q} \leq C_M(1 + |p| + |q|)^{-M}.$$

Taking $M > 2N$, we obtain

$$(4.72) \quad \sum_{p,q \in \mathbb{Z}^N} c_{p,q} < \infty.$$

Since G is finite, (4.71) and (4.72) imply

$$\begin{aligned} & \sum_{\sigma,\tau \in G} \sum_{p,q \in \mathbb{Z}^N} c_{p,q} \|A_{\sigma,\tau,p,q}(f)\|_{S^{N,\infty}(L^2(\mathbb{R}^N,dw))} \\ & \lesssim |G|^2 \left(\sum_{p,q \in \mathbb{Z}^N} c_{p,q} \right) \|f\|_{L^N(\mathbb{R}^N,dx)} \\ & \lesssim_{N,R,k} \|f\|_{L^N(\mathbb{R}^N,dx)}. \end{aligned}$$

Therefore the series

$$\sum_{\sigma,\tau \in G} \sum_{p,q \in \mathbb{Z}^N} c_{p,q} A_{\sigma,\tau,p,q}(f)$$

converges in $S^{N,\infty}(L^2(\mathbb{R}^N, dw))$, and

$$(4.73) \quad \left\| \sum_{\sigma, \tau \in G} \sum_{p, q \in \mathbb{Z}^N} c_{p,q} A_{\sigma, \tau, p, q}(f) \right\|_{S^{N,\infty}(L^2(\mathbb{R}^N, dw))} \lesssim_{N,R,k} \|f\|_{L^N(\mathbb{R}^N, dx)}.$$

It remains only to identify the operator represented by this series. By the definition of $A_{\sigma, \tau, p, q}(f)$, the integral kernel of its summand corresponding to $I \in \mathbb{D}_\Gamma$ is

$$\lambda_{\sigma, I}(f) e_{p, q, I}^{\sigma, \tau}(x) h_{p, q, I}^{\sigma, \tau}(y).$$

Hence the kernel identity (4.32) in Lemma 4.3 gives

$$M_f(-\Delta_{\text{Dunkl}})^{-1/2} = C \sum_{\sigma, \tau \in G} \sum_{p, q \in \mathbb{Z}^N} c_{p,q} A_{\sigma, \tau, p, q}(f).$$

More precisely, the two sides have the same integral kernel

$$f(x) K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(x, y)$$

for almost every (x, y) with $d(x, y) > 0$; hence they agree initially on the natural test-function class and therefore, by the boundedness of the right-hand side, as operators on $L^2(\mathbb{R}^N, dw)$.

Combining this identity with (4.73), we obtain

$$M_f(-\Delta_{\text{Dunkl}})^{-1/2} \in S^{N,\infty}(L^2(\mathbb{R}^N, dw))$$

and

$$\left\| M_f(-\Delta_{\text{Dunkl}})^{-1/2} \right\|_{S^{N,\infty}(L^2(\mathbb{R}^N, dw))} \lesssim_{N,R,k} \|f\|_{L^N(\mathbb{R}^N, dx)}.$$

Absorbing the structural constants into $C_{N,R,k}$ gives (4.70). $\square$

REFERENCES

- [1] J. Anker, J. Dziubański, and A. Hejna. Harmonic functions, conjugate harmonic functions and the Hardy space H^1 in the rational Dunkl setting. *J. Fourier Anal. Appl.*, 25(5):2356–2418, 2019. [15](#)
- [2] Z. Fan, M. Lacey, and J. Li. Schatten classes and commutators of Riesz transform on Heisenberg group and applications. *J. Fourier Anal. Appl.*, 29(2):Paper No. 17, 28, 2023. [6](#)
- [3] Z. Fan, M. Lacey, J. Li, M.N. Vempati, and B.D. Wick. Besov space, Schatten classes, and commutators of Riesz transforms associated with the Neumann Laplacian. *Michigan Math. J.*, 75(1):199–224, 2025. [6](#)
- [4] Z. Fan, M.T. Lacey, J. Li, and X. Xiong. Schatten–Lorentz characterization of Riesz transform commutator associated with Bessel operators. Preprint available at <https://arxiv.org/abs/2403.08249>, 2024. [6](#)
- [5] Y. Han, M. Lee, J. Li, and B.D. Wick. Riesz transforms and commutators in the Dunkl setting. *Anal. Math. Phys.*, 14(3):Paper No. 46, 32, 2024. [3](#)
- [6] G. Heckman. Coxeter groups. Lecture notes available at <https://www.math.ru.nl/~heckman/CoxeterGroups.pdf>. Radboud University Nijmegen, 2018. [7](#), [8](#), [10](#)
- [7] S. Helgason. *Differential geometry, Lie groups, and symmetric spaces*, volume 80 of *Pure and Applied Mathematics*. Academic Press, Inc. [Harcourt Brace Jovanovich, Publishers], New York-London, 1978. [3](#)
- [8] T. Hytonen. Schatten properties of commutators on metric spaces. Preprint available at <https://arxiv.org/abs/2411.02613>, 2024. [6](#), [7](#)
- [9] M.T. Lacey, J. Li, and B.D. Wick. Schatten classes and commutator in the two weight setting, I. Hilbert transform. *Potential Anal.*, 60(2):875–894, 2024. [6](#)
- [10] J. Li, J. Pipher, and L.A. Ward. Dyadic structure theorems for multiparameter function spaces. *Rev. Mat. Iberoam.*, 31(3):767–797, 2015. [5](#)
- [11] J. Li, X. Xiong, and F. Yang. Schatten properties of Calderón–Zygmund singular integral commutator on stratified Lie groups. *J. Math. Pures Appl. (9)*, 188:73–113, 2024. [6](#)

- [12] R. Rochberg and S. Semmes. Nearly weakly orthonormal sequences, singular value estimates, and Calderon-Zygmund operators. *J. Funct. Anal.*, 86(2):237–306, 1989. 6, 7
- [13] Z. Wei and H. Zhang. Complex median method and Schatten class membership of commutators. Preprint available at <https://arxiv.org/abs/2411.05810>, 2024. 6
- [14] Z. Wei and H. Zhang. On the boundedness and Schatten class property of noncommutative martingale paraproducts and operator-valued commutators. Preprint available at <https://arxiv.org/abs/2407.14988>, 2024. 6

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